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Preliminary Design of PRIMAVERA: Preliminary Investigation Mission to Achieve Venusian Reconnaissance in Atmosphere

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Abstract

PRIMAVERA, or the Preliminary Investigation Mission to Achieve Venusian Reconnaissance in Atmosphere, is a Phase 0/A study for a spacecraft developed by thirty students during the week-long 2024 Concurrent Engineering Workshop (CEW) hosted by the European Space Agency (ESA) in Transinne, Belgium. The mission objective was to develop a spacecraft to deliver a floating platform payload into the Venusian atmosphere where it would take in situ measurements in order to study the Venusian clouds. PRIMAVERA would be launched on an Ariane 64, and a gradual Hohmann transfer was planned to take the single-thruster spacecraft to its final circular, equatorial Venusian orbit at 5000 km. There, the payload consisting of a gondola with instrumentation attached to a helium variable-altitude balloon would descend to its operational altitude between 50 and 65 km. An iterative approach was taken towards systems design while following concurrent engineering principles. Students from ESA member states, Canada, and Slovakia were divided into teams of two to four members for each of the following subsystems: structures, configuration, thermal, power, communications and data handling, attitude and orbital control systems, propulsion, trajectory analysis, ground segment, and cost engineering. Subsystems were worked on simultaneously and parameters were tracked using ESA's Concurrent Model-based Engineering Tool (COMET software). This paper details the results of the preliminary design of PRIMAVERA.

Nomenclature

Trajectory

$F_{a,max}$ - Maximum occurring atmospheric drag
 ρ - Venusian atmospheric density
 h - Orbit altitude
 $(C_d A)_{max}$ - Maximum satellite drag area
 v - Orbit velocity
 A_{max} - Satellite reference area
 S_{Earth} - Earth solar constant
 S_{Venus} - Venus solar constant
 $F_{s,max}$ - Maximum radiation drag
 c - Speed of light
 m - Satellite mass
 Δv_d - Target orbit ΔV discrepancy/perturbation

Propulsion

I_{SP} - Specific Impulse
 $MON-3$ - Mixed oxides of nitrogen; 3% nitric oxide
 MMH - Monomethylhydrazine

Power

$GaAs$ - Gallium Arsenide

Attitude & Orbital Control Subsystems (AOCS)

T_g - Gravity gradient torque
 μ_{Venus} - Venus' gravitational parameter
 R - Orbital radius
 I_z, I_y - Total moments of inertia in z, y-directions
 θ - Maximum deviation from vertical
 T_{srp} - Torque arising from solar radiation pressure
 F - Force provided by solar radiation
 c_{ps} - Spacecraft centre of pressure
 c_g - Spacecraft centre of gravity F_s - Solar flux
 c - Speed of light
 A_s - Exposed spacecraft surface area
 ρ - Reflectivity of spacecraft surface
 i - Sunlight incident angle
 $T_{slew,max}$ - Maximum required slewing torque
 $\theta_{slew,max}$ - Maximum angle required to slew through
 I_{max} - Maximum moment of inertia
 $t_{slew,min}$ - Minimum slew manoeuvre time
 F_{RCS} - RCS thrust level
 T_{RCS} - Torque to be dumped via RCS
 t_{burn} - RCS burn time
 L - Reaction wheel stored angular momentum
 r_{RCS} - RCS thruster lever arm
 $m_{prop,RCS}$ - Required RCS propellant mass
 R_{Earth} - Earth radius

Structures & Mechanisms

f_{axial} - Axial natural frequency
 A_T - Cross-sectional area

E - Young's modulus

H - Spacecraft height

M - Spacecraft mass

$f_{lateral}$ - Lateral natural frequency

I_x, I_y - Total moments of inertia in x, y-directions

Thermal

Q_{out} - Total outgoing thermal load

Q_{IR} - Infrared radiative heat loss

Q_{diss} - Heat loss dissipated internally

Q_a - Heat absorbed from Venus albedo

Q_{Sun} - Heat from direct solar radiation

Acronyms/Abbreviations

Introduction & Concept

PRIMAVERA - Preliminary Investigation Mission to Achieve Venusian Reconnaissance in Atmosphere
CEW - Concurrent Engineering Workshop
ESA - European Space Agency
AOCS - Attitude and Orbital Control Subsystems
COMET - Concurrent Model-based Engineering Tool
CDF - Concurrent Design Facility
ESTEC - European Space Research and Technology Centre

Trajectory

VOI - Venus Orbital Insertion

Propulsion

ECSS - European Cooperation for Space Standardisation

Power

EPS - Electrical power subsystem
BOL - Beginning-of-life
Li-ion - Lithium-ion
Ni-H - Nickel-hydrogen
Ni-Cd - Nickel-cadmium
DoD - Depth of discharge
P-string - parallel string
PCDU - Power Conditioning and Distribution Unit

Communications & Data Handling

TT&C - Telemetry, Tracking and Command
TWTA - Travelling-Wave Tube Amplifier
RFDN - Radio Frequency Distribution Network
OBC - Onboard Computer

Attitude & Orbital Control Subsystems (AOCS)

COM - Communications
PROP - Propulsion
NASA - National Aeronautics and Space Administration
RCS - Reaction Control System
IMU - Inertial Measurement Unit
CSS - Coarse Sun Sensor
TRL - Technology Readiness Level
CONFIG - Configuration
FSS - Fine Sun Sensor
CoM - Centre of Mass

Ground Segment

ESTRACK - European Space Tracking
STK - Systems Tool Kit
FoV - Field Of View

Structures & Mechanisms

HDRM - Hold Down and Release Mechanism
SADA - Solar Array Drive Assembly

Thermal

IR - Infrared radiation MLI - Multi-Layer Insulation
OSR - Optical Solar Reflector

Configuration

LVA - Launch Vehicle Adapter
PLA6 - Payload Adapter for Ariane 6

Cost Engineering

USCM8 - Unmanned Space Vehicle Cost Model, Eighth Edition

Systems Engineering

CONOPS - Concept of Operations
LEOP - Launch and Early Operations
MAN - Manoeuvre Mode
COM_PL - Payload Communications Mode
COM_DTE - Earth Communications Mode
ECL - Eclipse Mode
EOL - End-of-Life

1. Introduction & Concept

PRIMAVERA, which stands for Preliminary Investigation Mission to Achieve Venusian Reconnaissance in Atmosphere, was a phase 0/A spacecraft mission study developed by students during the week-long 2024 Concurrent Engineering Workshop (CEW) at the European Space Agency (ESA) in Belgium, facilitated by ESA Academy (Figures 1, 2) under ESA's education department. Utilising concurrent engineering design philosophies and working with both proprietary ESA calculation materials and the COMET software package, as used in the ESTEC Concurrent Design Facility (CDF), thirty students from ESA member states, Canada, and Slovakia were divided into teams of two to four members for each of the following subsystems:

- Structures.
- Configuration.
- Thermal.
- Power.
- Communications & Data Handling.
- Attitude & Orbital Control Subsystems (AOCS).
- Propulsion.
- Trajectory Analysis.
- Ground Segment.
- Cost Engineering.

The mission statement for PRIMAVERA was "the mission shall deliver and operate a floating platform into the Venusian atmosphere to perform in situ scientific measurements within Venus' clouds." The two mission objectives were as follows:

1. Send a carrier spacecraft to Venus to deliver a floating platform into the Venusian atmosphere.
2. Perform in situ measurements to study the Venusian clouds.

The mission requirements are listed in Table 1. This mission was inspired by both the Venus Flagship Mission and EnVision. The design of the payload was beyond the scope of the project, however, the characteristics of the payload had implications on the mission requirements. The payload, called the floating platform, consists of a gondola containing instrumentation attached to a helium variable-altitude balloon. The balloon has no propulsion

system and achieves altitude control through moving helium between its zero-pressure balloon and its pressurised internal balloon. A preliminary concept of operations is shown in Figure 5. The payload specifications are listed in Table 2. The dimension and configuration of the payload is detailed in Figure 6. Figure 3 displays the mission patch that was developed for the mission throughout the week.



Fig. 1. Workshop Facilitator [1]

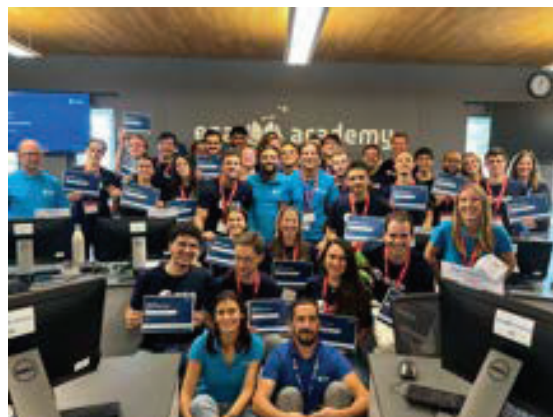


Fig. 2. Workshop Completion



Fig. 3. PRIMAVERA Mission Patch

2. Trajectory

An Ariane 6 in the 64 configuration is required to launch the combined spacecraft on a direct escape trajectory to Venus. The spacecraft pair has a dry mass of 2490 kg, and a total launch mass of 6350 kg. The primary launch window occurs in June 2032 with an arrival in August 2033 after a 440-day cruise. A secondary window in December 2032, which allows for an earlier arrival with a shorter 145-day cruise, could also require a spacecraft manoeuvre to supplement the launcher's second-stage escape burn. A direct escape trajectory was chosen over loitering in high Earth orbit in order to transfer a greater portion of the propulsive burden to the launcher.

Prior to Venus orbital insertion (VOI), the atmospheric probe separates and performs a direct hyperbolic-velocity entry near the equator. The orbiter executes a 920 m/s manoeuvre to capture a highly elliptical Venus orbit. The probe then drifts around Venus over four to five days due to the atmosphere's superrotation, maintaining a nearly fixed latitude. To maximise communication windows, an equatorial entry is targeted. The orbiter must also be placed in a near-equatorial orbit. This requirement would add a small increment to the circularisation manoeuvre, but with Venus' low obliquity, we find that the arrival inclination is already typically less than 1°.

The operational altitude of the orbiter was selected through a trade-off analysis. A higher orbit reduces the required circularisation burn, increases contact time with the probe, and decreases time spent in Venus' shadow. On the other hand, a lower orbit improves the communications signal-to-noise ratio with the probe. The selected altitude of 5000 km, corresponding to a 214 minute orbital period, enables communication with the probe for 17% of the time, in windows of around 40 minutes. The orbiter is eclipsed from the Sun approximately 18% of the time, also in 40-minute windows. Since the probe drifts throughout its motion, these communication and eclipse windows gradually shift in and out of phase with a ~4-day synodic cycle.

In order to reach the target orbit at an altitude of 5000 km around Venus with a gradual Hohmann transfer, a trade-off analysis between the total circularisation period and the individual perigee burn times was conducted. Visualised in Figure 7, the possible options are plotted with total duration of the manoeuvre on the y-axis and individual burn times on the x-axis. Opting for 11 burns at constant fuel flow per burn, the circularisation period is 32 days. This option was selected to keep the total circularisation time as low as possible, while still maintaining

margin towards the maximum thruster burn time of 20 minutes. This period further allows for functionality tests of both the onboard systems and the atmospheric probe before scientific data collection is started.

An estimation was performed to determine the magnitude of target orbit perturbations resulting from solar radiation and atmospheric drag. The maximum occurring atmospheric drag $F_{a,max}$ is determined using Equation 1:

$$F_{a,max} = -\frac{1}{2}\rho(C_d A)_{max}v^2 \quad (1)$$

Where ρ is the Venusian atmospheric density at the orbit altitude $h = 5000$ km, and the maximum satellite drag area $(C_d A)_{max}$ is perpendicular to the orbit velocity v . The circular orbital velocity at the orbit altitude calculates to $v = 5.422$ km/s. The drag coefficient C_d is set to a conservative value of 2.3, which is in line with some existing spacecraft according to Gargiulo et al. [2]. The maximum satellite reference area A_{max} is the sum of the cuboid body reference area (9 m²) and solar panel reference area (16.7 m²), when perpendicular to the orbital velocity. Finally, atmospheric data is taken from a NASA Venus atmospheric model [3]. The logarithmic model specifies a density of $\rho = 8.23 \times 10^{-14}$ kg/m³ at an altitude of 450 km for high density and maximum solar activity. Therefore, using this density value at an altitude of 5000 km is highly conservative, but acts as a pre-design estimation. After insertion of all parameters, the conservative maximum atmospheric drag calculates to $F_{a,max} = -7.15 \times 10^{-5}$ N.

Considering now the solar radiation pressure, the solar constant S is approximately 1.91 times higher at Venus than at Earth due to the quadratic distance relationship. Taking S_{Earth} from Johnson [4] results in a value of $S_{Venus} = 2664$ W/m². The maximum radiation drag can be expressed as:

$$F_{s,max} = -\frac{S_{Venus}A_{max}}{c} \quad (2)$$

with the speed of light c . The resulting solar force is thus $F_{s,max} = -2.3 \times 10^{-4}$ N. The drag forces are assumed to be their maximum value for the whole mission duration of one Earth year. After integration and considering a satellite mass of $m = 3000$ kg, the resulting $\Delta v_d = 3.2$ m/s. This value is low enough to not consider any orbital correction manoeuvres.

3. Propulsion

3.1 Propulsion Subsystem Requirements

The propulsion system was designed to meet the requirements of both the baseline launch window in June 2032 and the contingency window in December 2032. These dates were determined by computing the optimal launch windows in the case of impulsive manoeuvres and the choice between impulsive and continuous manoeuvres. Consequently, the definition of a propulsion system capable of delivering said manoeuvre type was made after a careful trade-off analysis. The propulsion system must be able to deploy the payload at equatorial Venusian latitude and park the orbiter at a 5000 km circular orbit. It is required to be based on European technology and to thus be compliant with the European Cooperation for Space Standardisation (ECSS). Furthermore, the system had to be designed so that the total wet mass at launch is lower than the launcher's maximum capacity for a direct transfer (~6.6 tonnes).

3.2 Propulsion Subsystem Design

A fundamental trade-off analysis had to be conducted for the choice between chemical and electric propulsion. The main advantages of the former were the ability to exploit impulsive manoeuvres, high thrust, and simple architecture. The main advantage of the latter was the high specific impulse characteristic of electric propulsion. By performing preliminary manoeuvre analyses and accounting for the available launch windows, it was determined that an advantageous set of impulsive manoeuvres was possible, thanks to planetary positioning. This would allow for a faster arrival date and, most importantly, a lower overall mass. Consequently, chemical propulsion was chosen.

The chemical propulsion system was designed with an assumed maximum qualified filling ratio of the tanks at 95% of the nominal volume. In the conceptual modelling, the propellant densities were assumed to be at standard conditions and the pressurisation was modelled as isothermal. The system was designed to maximise I_{SP} and minimise the total mass of the craft. Thereby, due to the large mass of the spacecraft, a thruster with a relatively high thrust was chosen in order to reduce gravity losses. For system simplicity and to minimise the required mass and volume, the secondary thrusters for the AOCS were chosen to be mono-propellant thrusters sharing a propellant reservoir with the main thruster.

A specific evaluation was carried out regarding the number of main thrusters. Since the mass of a single thruster of appropriate size was far lower than the total

mass of the spacecraft, a two-thruster configuration seemed tempting to raise the thrust-to-mass ratio and to grant redundancy. However, to prevent issues with thrust misalignment and to keep the architecture simple, a configuration with a single main thruster was preferred.

In Figure 8, a simple schematic of the suggested propulsion system is shown. The chosen design minimises the number of tanks and permits a symmetrical configuration, also due to an almost isovolumetric O/F ratio of the chosen propellant combination.

A preliminary component choice was made to demonstrate the feasibility and realism of the system while also providing a first iteration model. The main thruster is the LEROS 4 (Fig. 9) from Nammo AS running on MON-3/MMH, delivering an I_{SP} of 321 s and a nominal thrust of 1 kN at a comparably low mass of 8.41 kg [5]. As propellant tanks, two PTG-1908 tanks [6] from MT Aerospace AG were selected, each having a volume of 1908 L, which resulted in a close to optimal filling ratio. For the Helium pressurisation tanks, four PVG-120 tanks [7] from MT Aerospace AG with a volume of 120 L were chosen. The equipment was specifically chosen in alignment with the expected thermal ranges and based on the required pressure ratings.

3.3 Open Points

Future work should focus on a more sophisticated model for the tank pressurisation, so as to use a more realistic adiabatic model for manoeuvres. In addition to this, the extended investigation of alternative propulsive configurations for further system optimisation is required, alongside a more detailed mathematical model to account for gravity losses — the current one is simple, but overly conservative.

4. Power

The electrical power subsystem (EPS) must provide a continuous supply of electrical power in all mission modes and throughout all mission phases from the Launch and Early Operations Phase (LEOP) to End-of-Life (EOL). In addition to generating and storing electrical power, the subsystem must control and distribute electrical power to the spacecraft's components on regulated direct current (DC) power buses.

4.1 Mission Power Requirements

To identify the sizing cases for the EPS, each mission mode was analysed individually. Table 3 summarises the peak power requirements per mode, including a 20% system margin consistent with early-phase design practices.

Due to differing solar irradiance at Earth and Venus, sizing analyses were performed separately. Near Earth, battery sizing is driven by LEOP, while solar array sizing is constrained by the high demand of MAN, assuming continuous sunlight. Near Venus, Eclipse (ECL) mode dictates battery sizing and indirectly constrains array sizing. Comparative assessment shows ECL at Venus imposes the highest combined battery and array requirements, as arrays must sustain high-demand modes (e.g. COM_DTE) while recharging batteries for eclipse periods.

4.2 Solar Array Sizing & Design

Solar cell efficiency degrades significantly in the proximity of Venus due to elevated radiation and temperature levels. Therefore, high base efficiency is essential. Multi-junction gallium arsenide (GaAs) cells are identified as the most suitable option. Triple-junction GaAs cells were selected due to a favourable trade-off: a 7% cost increase over dual-junction cells yields a 22% increase in efficiency. The selected cells are comparable to the Azur Space 3G30C model, featuring a nominal cell voltage of 2.4 V and a beginning-of-life (BOL) efficiency of 26.8%. Each parallel string (p-string) consists of 12 cells in series, generating a bus voltage of 28 V. The design accounts for one p-string failure. At Venus orbit, solar irradiance is 2613 W/m². The worst-case sun incidence angle is 12.7°, accounting for Venus' axial tilt (2.7°) and a 10° pointing mechanism error in an equatorial orbit. Radiation exposure is expected to reduce solar cell efficiency to 20.8%. An additional thermal degradation of 3% per year is assumed, yielding an EOL efficiency of 17.8%.

Transmission efficiency is assumed to be 80% during exposure to sunlight and 60% during eclipse. The orbital period is 213.5 minutes, with an eclipse duration of 39.4 minutes. Based on EOL conditions, the required solar cell area is 6.21 m². To manage thermal loads, a low packing factor is used, allowing for the integration of reflective surfaces to lower cell temperatures. This results in an effective deployed solar array area of 10.9 m². The array configuration consists of two deployable panels per spacecraft side. Each panel measures approximately 1.5 m × 1.82 m to align with spacecraft body constraints. Body-mounted arrays were not selected, as their pointing requirements for optimal power generation would conflict with those of communication modes (COM_DTE and COM_PL).

4.3 Energy Storage System

Following a trade-off analysis, lithium-ion (Li-ion) battery technology was selected over nickel-hydrogen (Ni-H) and nickel-cadmium (Ni-Cd) alternatives. The Li-ion solution offers substantial mass savings (52 kg compared

to Ni-H and 630 kg compared to Ni-Cd) while improving energy storage efficiency from 70% to 98%. Although Li-ion cells have narrower operational temperature ranges, the mass and efficiency benefits outweigh the thermal restrictions. The optimal depth of discharge (DoD) for Li-ion in this mission context is approximately 40%, balancing performance and lifetime. For the worst-case eclipse duration of 39.4 minutes and a peak power requirement of 4.7 kW, with 98% transmission efficiency, the total energy storage requirement is 7.86 kWh. At a bus voltage of 28 V, this corresponds to a required capacity of 281 Ah. Accounting for the failure of one p-string, the battery architecture consists of 8s142p cells. These are split into two battery units, each configured as 8s71p. Each p-string provides 2 Ah. The design is comparable to commercially available Enersys 8s modules. The resulting DoD is 34.1% for the provided configuration, indicating further potential for optimisation and weight saving in future design phases.

4.4 Power Conditioning & Distribution

An upsized variant of the Airbus VEGA-C Power Conditioning and Distribution Unit (PCDU) is selected. It supports a maximum output power of 5 kW and provides a regulated 28 V bus voltage. The unit is selected due to its compatibility with Li-ion battery systems and heritage in comparable missions. However, this specific upsized configuration is not yet flight-proven. Consequently, conservative mass and cost margins are applied.

4.5 Open Points

Further investigation is required to refine the PCPU selection and validate the design for the mission's specific operational environment. Opportunities for mass and cost optimisation remain.

5. Communications & Data Handling (C&DH)

5.1 Communications Subsystem Requirements

For the PRIMAVERA mission, the communications and data handling subsystem will need to close two links: the orbiter to probe communication (data link) and the orbiter to Earth communication (report link). These objectives must be met within the bounds of the requirements listed in Table 4. From simulations and analysis of the mission as a whole, parameters were calculated with input from all sub-teams to define the required capabilities of the system. These parameters are as follows:

- 5000 km altitude.
- Orbiter-Probe contact time is 38.4 minutes per 220 minute period.

- Orbiter-Earth contact time is 116 minutes per 610 minute period.
- S, X, and K_a bands are available for communication.
- Both links must be closed with a link margin above 3 dB.

5.2 Communications Subsystem Design

The probe must transfer all data generated per cycle to the orbiter. Over this 220-minute period, data is generated at 135 kbps, resulting in a total data size of 222.75 MB. Accounting for margins, this requires an approximate 800 kbps uplink speed to send all generated data to the orbiter. Once the data has been transferred to the orbiter, a compression algorithm is used to reduce the data payload being sent to Earth.

The main trade-offs relevant to this subsystem were: altitude vs power, link budgets vs power, contact times vs data rates, and contact frequency vs slew fuel. Higher altitude would increase the power requirements, however it would also allow for a longer contact window. This would in turn reduce the slew fuel required as the satellite would need to re-align with the probe less frequently. Tracking within the contact window itself was done with a gimbal attached to the receiving antenna - slew fuel was only needed from the end of a contact window to the start of a new one, where the probe approaches from the original direction.

One of the more significant trade-offs was contact time vs slew fuel — the number of full length contacts that were expected to be obtained for the data link and report link was directly decided by the amount of fuel that could be brought on board, thus affecting the mass of both the orbiter and the launcher. Other unique problems were the balance of the dual-link, ensuring data is sent to Earth within the defined time-frame, and the nature of the Venusian atmosphere, which heavily impacts transmission losses. A delicate balance had to be struck within the requirements to allow for sufficient link closure whilst maintaining a feasible launch mass.

An example of a trade-off and subsequent analysis is the data link communication band selection. The bounds of the analysis were that the altitude needed to be above 500 km, and the link budget had to be closed with a minimum margin of 3 dB. A lower data rate was desirable to reduce power requirements and signalling complexity while maintaining this link budget. An analysis for 3.5 GHz (S-band) at 20 W is shown in Figure 10.

Several communications components were initially selected. The data link is to be in the S-band, and thus uses the MGA 20 dB S-band horn antenna from A-INFO INC [8], coupled with the Lime Microsystems S-band transceiver [9]. The report link will have 2 antennas, the TRYO K_a-band antenna for reporting data to Earth, and the Beyond Gravity X-band helical antenna for receiving Telemetry, Tracking and Command (TT&C) from Earth. The former is a high-gain antenna (HGA) allowing for reliable transmission at higher data rates, while the latter is a low-gain antenna (LGA), relying on the power budget from Earth to provide a higher gain signal. Alongside these, the travelling-wave tube amplifier (TWTA) and the radio frequency distribution network (RFDN) were chosen to be the same as the EnVision Mission [10]. Moreover, the Airbus OSCAR onboard computer (OBC) was selected due to its high processing power and substantial storage capacity, as well as heritage flight performance. Given its proven reliability within space environments and seamless integration with other Airbus spacecraft systems, this ensures that it can perform effectively while managing power efficiently throughout the mission [11].

6. Attitude & Orbital Control Subsystems (AOCS)

6.1 AOCS Requirements & Drivers

The PRIMAVERA orbiter requires comprehensive AOCS to determine and control its attitude throughout the duration of the mission lifetime in order to capture, interpret and control attitude and positioning data, in addition to monitoring all Δv manoeuvres required across the mission profile. In the case of PRIMAVERA, which will endure a variety of disturbance environments between Earth and Venus, dynamic analysis can be quite complex. Moreover, the AOCS architecture is required to handle any unique profiles outside of the mission brief, due to any scenario ranging from minor equipment failure to collisions with foreign bodies. The main AOCS design drivers (Table 5) arise from the worst-case pointing requirements for the communications subsystem, particularly maintaining Earth-pointing of the HGA. The onboard COM system acts as relay in a store and forward architecture whereby data is initially uplinked from the floating platform within the Venusian atmosphere to the orbiter during contact periods. The stored data may then be subsequently downlinked to the Earth. The spacecraft has only one HGA which is not mechanically steered. As a result, the spacecraft itself must slew between Venusian nadir and Earth-pointing attitudes. These requirements, along with reaction wheel off-loading constraints, led to close coordination between the AOCS, communications, and propulsion teams. Based on these drivers and high-level design assumptions (Table 6), subsystem

requirements were defined (Table 7).

6.2 AOCS Design

When laying out the design for PRIMAVERA's AOCS equipment, a multi-faceted and iterative approach was adopted in line with that of the COMET software package (and as such, similar to that adopted across the remaining additional subsystems). The main iteration steps are displayed in Table 8, as outlined by guidance from both ESA and NASA's Goddard Space Flight Centre. Furthermore, control modes are defined in Table 9.

6.2.1 Initial Analysis & Trade-Off Study

To meet the mission's high-precision pointing needs, baseline three-axis stabilisation was selected, requiring at least two reaction control system (RCS) thruster quad units (eight nozzles). To ensure fault tolerance, an additional two quads were added (thus, sixteen nozzles total), providing redundancy and full control authority even after failure. This configuration allows for pure torque and translational manoeuvres while minimising parasitic Δv .

Reaction wheels are used as the primary attitude actuators due to their agility and efficiency, especially in the absence of a main engine with gimbal capability. A shared fuel reservoir between the main engine and RCS assists in minimising mass and cost. Momentum unloading, based on EnVision mission data [12], is expected once per day (~ 0.3 mm/s ΔV), staying below 0.07 m/s over a full Venusian year. For attitude sensing, star trackers are the primary sensors, offering full three-axis determination. These are supported by gyroscopes and accelerometers (forming an IMU) to improve performance during transients, with CSS units included for SAFE mode due to their high technology readiness level (TRL) and reliability.

6.3 AOCS Equipment Sizing & Configuration Selection

6.3.1 Rotational Dynamics Analysis

To size AOCS hardware, key environmental torques were estimated. Magnetic and aerodynamic effects were neglected due to their complexity, focusing instead on gravity gradient and solar radiation torques (Equations 3, 4). Using spacecraft inertia data (Table 6) and dimensions from the Configuration team (2.1 x 3 x 3 m), torque values were derived (Table 10). Total disturbance torque was computed via root-sum-square, and maximum slewing torque via Equation 6.

$$T_g = \frac{3\mu_{Venus}}{2R^3} |I_z - I_y| \sin 2\theta \quad (3)$$

$$T_{srp} = F(c_{ps} - c_g) \quad (4)$$

Where:

$$F = \frac{F_s}{c} A_s (1 + \rho) \cos i \quad (5)$$

$$T_{slew,max} = \frac{4\theta_{slew,max} I_{max} \pi}{180 t_{slew,min}^2} \quad (6)$$

These values are processed using proprietary ESA tools to extract peak torque and momentum over an orbit (Table 11), with components separated into cyclical and secular trends via Fourier-based analysis.

6.4 Reaction Wheel Sizing

Reaction wheel sizing for PRIMAVERA considers a 20-minute maximum manoeuvre window required by the communications subsystem for reliable uplink/downlink. Although the AOCS loadout estimates a typical manoeuvre time of 12 minutes, the 20-minute worst-case (1200 s) is used for design. A time-optimal 'bang-bang' slew profile is assumed (Figure 11), applying maximum torque for 600 s, then reversing it for the remaining 600 s to bring the spacecraft to rest. This control method, known for its simplicity and suitability for systems with limited computational resources, is ideal for rapid slewing under gravity gradient influences. While efficient, it may induce vibration or overshoot, requiring RCS corrections and additional momentum dumps. The required angular momentum is calculated as the product of slew torque and half the manoeuvre duration, yielding 49.2 Nms. Based on this, the Collins Aerospace RSI 68-75/60 reaction wheel (Figure 12, 68 Nms capacity) is selected (Table 12). This actuator selection suits the spacecraft's limited volume, offering 50% and 13% safety margins for torque and momentum, respectively. A tetrahedral reaction wheel configuration enhances control by providing three-axis redundancy and fault tolerance, maintaining functionality after a single wheel failure. This specific geometry is implemented to amplify the effective momentum and torque per axis by a factor of 1.63, leveraging the geometric arrangement to ensure efficient performance and a safety margin against disturbances. Optimisation based on unequal inertias (I_{xx} , I_{yy} , I_{zz}) is also possible, ensuring robust performance under disturbance conditions.

6.5 RCS Thruster Sizing

The primary design driver for PRIMAVERA's AOCS equipment is the need for frequent large-angle repointing manoeuvres between Earth-pointing and Venus-zenith orientations, stemming from its store-and-forward communications architecture. Data is uplinked from the

atmospheric platform using a steerable S-band antenna, temporarily stored onboard, and subsequently downlinked to Earth via a fixed K_a-band high-gain antenna. Based on expected data volumes, the orbiter must execute approximately 2.79 downlinks per day. To accommodate worst-case scenarios, a 20-minute window is reserved before and after each downlink for attitude slews. The most demanding scenario involves a 160° in-plane slew from Venus zenith to Earth-pointing, based on Venus-Earth orbital geometry. Assuming a time-optimal bang-bang control profile, this results in a maximum reaction wheel torque requirement of 0.082 Nm and a total angular momentum of 97.68 Nms accumulated over the 20-minute manoeuvre (600 s torque + 600 s reverse). However, the reaction wheels do not absorb the full 97.68 Nms. Following the downlink, the spacecraft must return to Venus-zenith orientation to re-establish uplink to the atmospheric platform, but during the downlink period, PRIMAVERA continues orbiting Venus. As a result, the return slew angle is less than the initial 160°, reducing the net momentum change required.

By proportionally scaling the angular momentum to the estimated return angle (from orbital mechanics: ~134°), the maximum net angular momentum that must be dumped by the RCS thrusters is approximately 81.8 Nms (Figure 13). This is the difference between the two slews - outbound and return - and forms the worst-case input for momentum offloading requirements. The average thrust level required per nozzle is derived via Equation 7, based on this value and assuming a symmetrical quad-thruster configuration. While reaction wheels remain the primary actuators for precision pointing and slewing, the RCS system is essential for momentum management, detumbling, and disturbance rejection — ensuring robust, fault-tolerant control across all mission phases.

$$F_{RCS} = \frac{T_{RCS}}{4t_{burn}} \quad (7)$$

Where:

$$T_{RCS} = \frac{L}{r_{RCS}} \quad (8)$$

Using the maximum angular momentum of 81.8 Nms and a conservative 3 s burn duration, each RCS thruster is sized to deliver 4.5 N of thrust. To reduce mass and complexity, the RCS system shares a Hydrazine monopropellant tank with the main engine. Based on mission operations and a trade-off study with the communications team, an average of 2.75 slews per day is expected, with one momentum dump daily. This results in a total RCS burn time of approximately 675 s over

a 225-day Venusian year, corresponding to a nominal propellant mass of 3.5 kg, found via Equation 9. To accommodate a one-year mission extension and add a 40% safety margin, the final requirement is 10 kg - well within the 20 kg limit allocated by PROP. The Northrop Grumman MRE-1.0 monopropellant thruster (Table 13) is thus selected based on its compact design, high thrust efficiency, and strong heritage. It provides rapid response for detumbling, momentum unloading, and fine attitude control. Its broad operating pressure range and high cycle life support robust performance across all nominal and contingency scenarios. The selected redundancy rationale allows for full authority over all three axes in all likely contingency scenarios.

$$m_{prop,RCS} = \frac{A}{B} \quad (9)$$

Where:

$$A = \frac{2F_{RCS}t_{burn,total}}{I_{SP}} \quad (10)$$

$$B = \frac{\mu_{Venus}}{1000R_{Earth}^2} \quad (11)$$

6.6 Additional AOCS Equipment Selection

6.6.1 Star Trackers

The ST200 Star Tracker from AAC Clyde Space (Figure 14) was selected due to its strong performance under high-albedo conditions, addressing Venus' 0.69 reflectivity (0.43 for Earth) by facing away from the planet and Sun to minimise optical and thermal interference. Offering 30 arcsec pitch/yaw and 200 arcsec roll accuracy, it exceeds the required 0.133°/s slew rate, while its compact form factor, low power consumption, and passive redundancy make it well-suited to PRIMAVERA's constraints (Table 14). Proven flight heritage further validates its selection.

6.6.2 Sun Sensors

Coarse sun sensor (CSS) units are essential for safe mode operations and backup attitude determination in case of primary sensor failure. Fine sun sensors (FSS) were excluded due to added complexity, mass, and cost, which were unnecessary given the robust AOCS design. An analog CSS unit from Bradford Space (Figure 15) was selected for its low power usage, compact form, and proven reliability. It meets the 1.5° pointing accuracy requirement and performs well under Venus' high-albedo conditions. Its flight heritage, including EnVision, along with passive redundancy and durability, confirms its suitability for the mission (Table 15).

6.6.3 Gyroscopes & Accelerometers

The ASTRIX 1000 gyroscope (Figure 16; Airbus Defence and Space) and AI-Q-2030 accelerometer (Figure 17; InnaLabs) were jointly selected to form a high-performance IMU suited to PRIMAVERA's mission environment. The ASTRIX 1000 offers excellent angular rate stability ($0.01^\circ/\text{h}$), a compact 3-axis design, low power usage, and active redundancy — proven in telecom satellite applications (Table 16). Paired with the AI-Q-2030 (Table 17), known for its precision, low mass, and integration ease, the result is a reliable, compact IMU with strong flight heritage, capable of meeting the mission's precision and durability demands.

6.7 Open Points

Despite the work that has been carried out, several open points remain. Central to these further objectives is the refinement of the pointing strategy and worst-case slew dynamics, requiring precise alignment with the spacecraft's centre of mass (CoM) in order to mitigate rotational instabilities and enhance attitude control precision. Additionally, a further strategic optimisation of RCS propellant mass through further iterations is essential for Δv budgets and extending operational longevity, although this would require leveraging advanced propulsion efficiency models. Moreover, there remains reasonable accommodation for further optimisation requests from interdisciplinary teams, underpinning the iterative and concurrent methodology with which the PRIMAVERA spacecraft was developed. This enables further enhancements in structural, thermal and navigational performance parameters, thereby bolstering the mission's relative robustness under the stringent environmental constraints observed across the lifetime of the spacecraft.

7. Ground Segment

The ground segment subsystem is responsible for the selection of a ground station (or ground stations) on Earth capable of closing the link with the orbiter for both payload data and TT&C.

7.1 Ground Segment Subsystem Requirements

A balance needed to be struck with the orbiter's altitude to maximise the strength of the orbiter's link with the floating platform while allowing a long enough window to downlink payload data back to Earth. Further considerations included reducing the size and power draw of the orbiter's antenna, the AOCS fuel restrictions for manoeuvring between pointing at the payload and the Earth, and the propulsive system and trajectory needed to reach the final orbit. To facilitate the decision, it was assumed that the orbiter could only downlink in daylight, and that the

orbiter must uplink from the payload every orbit. Additionally, the mission requirements specified that a European ground station (ESTRACK network) must be used (MIS-R-080), and that all payload data from the floating platform must be relayed to Earth within a given time-frame (MIS-R-040 and MIS-R-050). These requirements are from the mission requirements in Table 1.

7.2 Ground Station Selection

The following four factors were considered in the selection: number and frequency of passes, access time overlaps, eclipse durations affecting link visibility, and the ability to close the link budget. Some stations (e.g., Kiruna) were discarded due to infrequent passes, while others were eliminated for not supporting the right bands or dish size to achieve the required link margin (e.g., Redu). After an initial round of selection, a trade-off analysis was conducted between the ESTRACK stations at Cebreros, Spain, New Norcia, Australia, and Malargüe, Argentina, using access time simulations carried out within Ansys STK, the results of which can be seen in Figure 18.

Based on these results, the ESTRACK stations in New Norcia, Australia and Cebreros, Spain were chosen, which allow K_a-band downlink for payload data and both X-band up and downlink for TT&C. Out of all the ESTRACK stations with 35 m dishes and these bands, New Norcia (Figure 19) and Cebreros (Figure 20) gave the longest combined contact time with the orbiter over the course of an entire Earth year. Choosing the most capable stations for the required bands lessened the load for the orbiter's communications subsystem, and it remained within budget to operate two ground stations.

Visibility with the orbiter ranged from 7 to 13 hours per day, but solar eclipses and AOCS manoeuvres reduced usable downlink time to ~110 minutes per pass. This was still sufficient to support the expected science data rate of 600 kbps and total data volume per orbit. This setup enabled up to ~35% total coverage over one year of science operations, fulfilling the mission requirement MIS-R-040.

7.3 Open Points

In the future, more comprehensive simulations would need to be done taking into account the exact details of the spacecraft's antenna, such as the field of view (FoV), in order to iron out a more accurate schedule of operations. More in-depth analysis on these simulation results should also be conducted.

8. Structures & Mechanisms

8.1 Structural Subsystem Requirements

The structural requirements define the constraints that the spacecraft must satisfy to ensure compatibility with the launch vehicle, robustness under mechanical loads, and feasibility of integration. These requirements address stiffness, geometry, mass distribution, and interface compliance with Ariane 6. Table 18 summarises the main structural requirements considered in the design.

8.2 Structural Subsystem Design

To determine the most suitable vehicle shape, the team first investigated the forms used in previous missions, comparing quadratic and cylindrical configurations. A quadratic shape was ultimately chosen, not only because it had been the most common solution historically, but also because it provides sufficient external surface area to accommodate the attachment of critical components such as solar panels and antennas. Once the general shape was decided, its dimensions were defined by considering both the Ariane 6 fairing envelope and the results of a natural frequency study. The final geometry, measuring 3 m × 3 m × 2.1 m, allowed the structure to achieve an axial natural frequency of 132.69 Hz and a lateral frequency of 127.06 Hz, comfortably above the required 125 Hz limit. These frequencies were calculated from Equations 12 and 13.

$$f_{axial} = \frac{1}{4} \sqrt{\left(\frac{A_T}{100^2} \times \frac{E \times 10^9}{HM} \right)} \quad (12)$$

$$f_{lateral} = 0.56 \sqrt{\left(\frac{E \times 10^9 \times \min(I_y, I_x)}{100^4 MH^3} \right)} \quad (13)$$

The structural layout was then refined in terms of materials and mass distribution. The skin was designed in glass fibre, chosen for its lightweight properties and adequate stiffness, with a total mass of 68 kg. The stringers, which provide reinforcement and rigidity to the overall structure, were produced in aluminium alloy, adding 188.16 kg. To complete the primary load-bearing frame, rings, frames, and fasteners were incorporated with an estimated mass of 60 kg. These elements together formed a robust yet efficient structure that could withstand the loads of launch while remaining compatible with the spacecraft's mass budget. Table 19 provides additional information on the structural components and Figure 21 shows the satellite structure.

For the internal subsystem supports, special attention was given to the propulsion tanks. The team opted for a two-part compression-based support system, able to hold the propellant and oxidiser tanks in place during

flight (Figure 22). This was built from an aluminium honeycomb sandwich with carbon-fibre reinforced plastic, a material previously used in other ESA missions due to its combination of low density and high resistance. The thruster structure (Figure 23), while simpler, follows the same philosophy of practicality and robustness, and is manufactured from aluminium alloy.

Finally, to connect the spacecraft to the launch vehicle, the Payload Adapter PLA6 1194 (Figure 24) was selected. This adapter, with a diameter of 1194 mm, ensures structural compatibility with Ariane 6 and integrates a low-shock payload separation system (LPSS). This solution minimises the mechanical disturbances experienced during spacecraft release while maintaining high reliability.

Overall, the structural design process followed a philosophy of simplicity, cost efficiency, and reliability. Lightweight materials, heritage-based solutions, and straightforward interfaces were consistently preferred, ensuring the spacecraft complied with all launcher requirements while also enabling ease of assembly and testing.

8.2.1 Mechanisms

The requirements of the spacecraft's mechanisms specify the functional needs for deployment, pointing, and adjustment of the spacecraft's moving subsystems. They ensure reliable operation during LEOP and guarantee that antennas, solar arrays, and deployable payloads function as intended. Table 20 presents the key requirements guiding the mechanisms design. The mechanisms were developed with the same philosophy of reliability and heritage, ensuring all moving parts could operate effectively during the mission. Each was selected or adapted from proven designs used in previous ESA missions, which reduced development risks while meeting PRIMAVERA's specific needs:

- **Antenna Pointing:** Implemented with the 33 Biaxial Gimbal MOOG (TRL 9), chosen for its low mass, full 360° rotation capability, and robustness, ensuring the orbiter antenna can follow the floating platform within the required 120° communication range (Figure 25).
- **Floating Platform Deployment:** Based on the system used in the ExoMars mission, but adapted for PRIMAVERA, this mechanism uses three pyro-nuts and inclined springs along helicoidal rails to eject the

platform with controlled spin (10–15 rpm) and velocity (0.3 m/s). Its total mass is 30 kg, in line with requirements.

- **High-Gain Antenna Bracket:** Inspired by EnVision, the HGA bracket provides a fixed mounting solution at a mass of around 6.5 kg, ensuring stability and minimising design complexity.
- **Solar Array Deployment:** For solar array deployment, ARQUIMEA's REACT hold down and release mechanism (HDRM) and hinges were chosen. These mechanisms, with heritage from the JUICE mission, provide low-shock, non-explosive deployment, are easy to reset during ground testing, and are capable of supporting the spacecraft's two-panel solar arrays, each weighing 31.5 kg (Figures 26, 28).
- **Solar Array Adjustment:** Once deployed, the solar arrays can be rotated up to 350° using MOOG's SADA, allowing for optimal orientation with respect to the Sun (Figure 27).
- **Yoke:** The arrays are structurally connected to the spacecraft through a stainless-steel yoke with a mass of 8.8 kg, ensuring robustness during both launch and deployment (Figure 29).

Through these carefully selected mechanisms, all functional requirements were satisfied, ensuring that the antenna, solar arrays, and payload platform can operate as intended in orbit while minimising risk through the use of established technologies. Table 21 summarises the quantity and the TLR of the mechanisms.

9. Thermal

The thermal control subsystem establishes and maintains the allowable temperature range of the spacecraft and its components by balancing internal heat generation with external heat inputs and radiative rejection. As a part of the preliminary design of the thermal subsystem on the PRIMAVERA mission, the design process defines requirements, temperature limits and environments, models hot and cold worst-cases and selects ideal control elements (insulation, radiators, heaters, transport paths, surface finishes), for the given environment. Analysis and tests are not within scope of this preliminary study.

9.1 Thermal Subsystem Requirements

The thermal subsystem's purpose is to maintain internal spacecraft temperature and temperatures of all subsystems within their operational temperature limits during all mission phases. During the preliminary design,

the requirements are based solely on operational temperature limits and survivable temperatures are left as a safety buffer. In future iterations, temperature limits can be modified such that survivable temperatures can also be taken into account during non-operational times. The thermal subsystem requirement is then defined: The thermal control subsystem shall keep the craft's internal temperature limit between -20°C to 45°C .

9.2 Design Assumptions

In order to streamline the design process, multiple assumptions are made beforehand to simplify and define the process:

- No payload-specific stability requirement. Scientific instrumentation does not require tight temperature stability or gradient control; bus-level thermal control is sufficient.
- External appendages are not driving. In the modelled Venus environment, external units (e.g., antennas) do not impose additional thermal constraints beyond the baseline finishes and radiator/heater architecture.
- Antenna-hosting face coverage. The platform face carrying the antenna is assumed fully covered by the antenna; its optical/thermal properties are taken as those of the antenna surface.
- Projected area approximation. For environmental loads — solar, albedo, planetary infrared radiation (IR) — the spacecraft projected area is approximated by the planet-facing side to bound external fluxes.
- Heater treated as system allocation. Electrical heating is treated as a platform-level allocation sized to keep the spacecraft above the lower limit (-20°C) in the cold case; no heater budgets are assigned to individual items at this stage.

9.3 Modelling Considerations

To choose and size the proper materials and components for the thermal subsystem, two boundary scenarios are defined where two worst environmental conditions are present. The analysis separates a steady-state hottest case and a transient coldest case. The cold case integrates the thermal capacitance with radiative losses and reduced dissipation during eclipse to size the heater power.

9.3.1 Hottest Case

In the hottest case scenario, all the external heat fluxes (solar, albedo, planetary IR) are at their highest. These inputs must be balanced against radiative emission by choosing the appropriate size and material of radiator (Figure 31). The equilibrium shown in Equation 14 must be met.

$$Q_{out} = Q_{IR} + Q_{diss} + Q_a + Q_{Sun} \quad (14)$$

9.3.2 Coldest Case

In the coldest case scenario, the vehicle is in eclipse and there is no heat flux from the Sun or albedo. All heat flux input is at its lowest (Figure 32). The equilibrium shown in Equation 15 must be met.

$$Q_{diss} = Q_{IR} + Q_{out} \quad (15)$$

9.4 Design Drivers

The design drivers which are essential for safe operation and maintenance over the whole spacecraft life cycle are the mission profile, spacecraft mass, geometry and insulation material, radiator material and solar array geometry, efficiency, cell material and surface finish (Figure 33). These have a basic implication on the heat mass flow to the spacecraft and its heaters, on the radiator surfaces, on solar array temperatures, and on the surface finish mass.

9.4.1 Insulating Material Analysis

According to the trade-off analysis conducted (Figure 34), the 1/2 mil Kapton insulating material is the best for the spacecraft's insulating purposes. Common key parameters for all considered materials were the mass, cost, power and the equilibrium temperature. The mass had the highest weight factor of 0.50, followed by the cost with 0.36, then by the power and the equilibrium temperature, with 0.07 weight factor for both.

9.4.2 Solar Array Reflector Analysis

A similar trade-off analysis (Figures 35, 36) was done for six material types for the solar arrays. The material types are divided into six different reflector categories, which are optical solar reflectors (OSRs), Z93 white paint, black paint, aluminised Kapton, metallic and miscellaneous reflectors. The selection criteria was the highest possible minimum temperature. The temperatures for all materials were calculated in accordance with the requirements and mission profile of the spacecraft and individual components. Key parameters for the trade-off were the mass, cost, and equilibrium temperature. Of all six reflectors, the most appropriate was the Z93 white paint reflector for the underside of the solar array, with an absorptivity of $\alpha = 0.17$ and an emissivity of $\epsilon = 0.92$. For the top side of the solar array, the triple-junction with $\alpha = 0.313$ and $\epsilon = 0.75$ and the mirror 5mil silvered Teflon with $\alpha = 0.05$ and $\epsilon = 0.78$ were chosen.

9.5 Preliminary Thermal Subsystem Design

9.5.1 Insulation

The spacecraft's insulation system utilises multi-layer insulation (MLI) made up of thin layers of Kapton, Kevlar, and Nylon to isolate the spacecraft from the harsh space environment. The insulation minimises heat absorption while ensuring efficient heat emission, helping maintain stable internal temperatures across extreme conditions. The number of layers was optimised based on trade-off analyses and previous missions, with 25 layers providing the desired performance. The oxidiser tank shall also utilise polyurethane insulation in order to keep its temperature above a limit of -15° (higher than that of -20° accepted for the platform). Tables 22 and 23 detail main body and individual component insulation respectively. Across individual components, and as can be expected, some have a lower temperature margin than others — stable internal temperatures under extreme conditions are ensured.

9.5.2 Radiating Surfaces

To dissipate excess heat, the spacecraft is equipped with OSR radiators. These radiators use quartz OSRs for their low absorptivity and high emissivity, ensuring effective heat rejection while absorbing minimal heat. The radiators are sized based on their thermal properties to maintain the spacecraft's temperature within operational limits. The chosen material effectively radiates the heat generated by the spacecraft's systems, helping maintain thermal balance across all mission phases. Radiating surface data is contained within Table 24.

9.5.3 Heat Distribution

The heat distribution system ensures that heat is evenly transferred throughout the spacecraft, preventing overheating in critical areas. This system includes heat pipes designed to transport excess heat to the radiators for dissipation. These pipes are highly conductive and filled with a working fluid that changes phase to carry heat away from hot components, ensuring a balanced temperature distribution. The selected heat pipes have been sized based on the spacecraft's thermal power production, as well as the required cooling capacity to maintain thermal equilibrium. Thermistors are used to monitor heat distribution levels, with heaters integrated to warm the spacecraft consistently during eclipse operations and keep systems above their limiting temperatures. As previously stated, white paint (Z93) is applied to the underside of the solar array to dissipate heat, thanks to its high emissivity and low absorptivity. Black paint, with very high absorptivity, is used on internal components to stabilise the spacecraft's temperature by transferring heat away through radiative means.

Tables 25, 26 and 27 summarise key data relating to the selected equipment for the heat distribution system.

9.6 Open Points

The following points require further attention:

- Finalise internal thermal link conductances in the nodal model.
- Define heater zoning and control set-points aligned with the internal layout.
- Refine total heater power via detailed transients and confirm margins.
- Optimise MLI layer count versus performance, mass, and integration breaks.
- Reassess heat-pipe length/mass by leveraging internal radiative coupling.

10. Configuration

10.1 Configuration Subsystem Requirements

The configuration team identified six subsystem requirements. These requirements are not constrained to this subsystem in particular, but affect the whole spacecraft. They can be found summarised in Table 28.

Compliance for the first two requirements was achieved by selecting the single-launch Ariane 64 long fairing, with the single Launch Vehicle Adapter (LVA) (see Figure 37). The third requirement was fulfilled by selecting a cubic shape, and by following designated design drivers. As the fuel tanks were the heaviest (and biggest) elements, it was decided to develop the whole spacecraft around them (see Figure 38). For the same reason (weight distribution), the reaction wheel assembly was situated as close as possible to the CoM, so that the moment of inertia was minimal. The arrangement of the equipment tray (see Figure 39), with respect to the communication equipment, was driven by the need to keep the different components of this subsystem as far as possible from sources of electromagnetic noise (such as the on-board computer), and the design of the thermal subsystem equipment (only to a high level) was driven by the need to keep it close to the radiator — in order to avoid inefficiencies related to the heat pipes and paths. The weight distribution of the whole spacecraft was chosen so that the CoM would be as fixed as possible along all mission phases. Moreover, the exterior disposition was selected following one of the trade-offs that will be explained later in the paper. The outcome of this trade-off is as follows: the balloon probe was stowed on face A (see Figure 40), and the ensemble of the main nozzle and

the Payload Adapter PLA6 1194 was installed on face B; the one interfacing with the rest of the launcher. Next, the X-band and K_a-band antennas were located on face C, opposite to face D, that carries the S-band antenna. Finally, the solar arrays were located on the free faces (E and F), equipped with a mechanism that allows for constant sun tracking, if needed. The location and the deployment mechanism of the solar arrays was designed so that there would be no obstacles inhibiting deployment or nominal operation.

The location of the big structures (stowed probe, solar arrays and PLA6) allows for an unobstructed FoV of the sensors and the antennas, therefore achieving compliance with the fourth requirement. In order to comply with the fifth requirement, the deployment of the solar arrays and the stowed probe was considered in the design. The location of the elements in faces A, E, and F does not interfere with either the release of the balloon probe or the deployment of the solar arrays. Similarly, the main nozzle was placed on a free face (face B), while the RCS thrusters were positioned on some of the edges of the cube. This arrangement ensured that their operation was not compromised by any obstruction, therefore achieving compliance with the sixth requirement.

10.2 Design Assumptions

Throughout the design process, four assumptions were determined and used:

1. In order to keep the design simple and less expensive, a box-like structure was used. This choice allowed for a 50% cheaper structure, according to the calculation sheets that were provided.
2. It was assumed that in the final configuration, the thrust vector would be passing through the CoM of the spacecraft.
3. Derived from one of the trade-offs (see later), it was assumed that the best position for the probe was face A (opposite to the main nozzle).
4. Finally, it was assumed that the spacecraft would comply with the launcher CoM requirements.

10.3 Analysis and Trade-offs

Two different configuration trade-offs were studied:

- Which face should be the one attached to the payload adapter?
- Should the main propulsion subsystem be jettisoned after VOI?

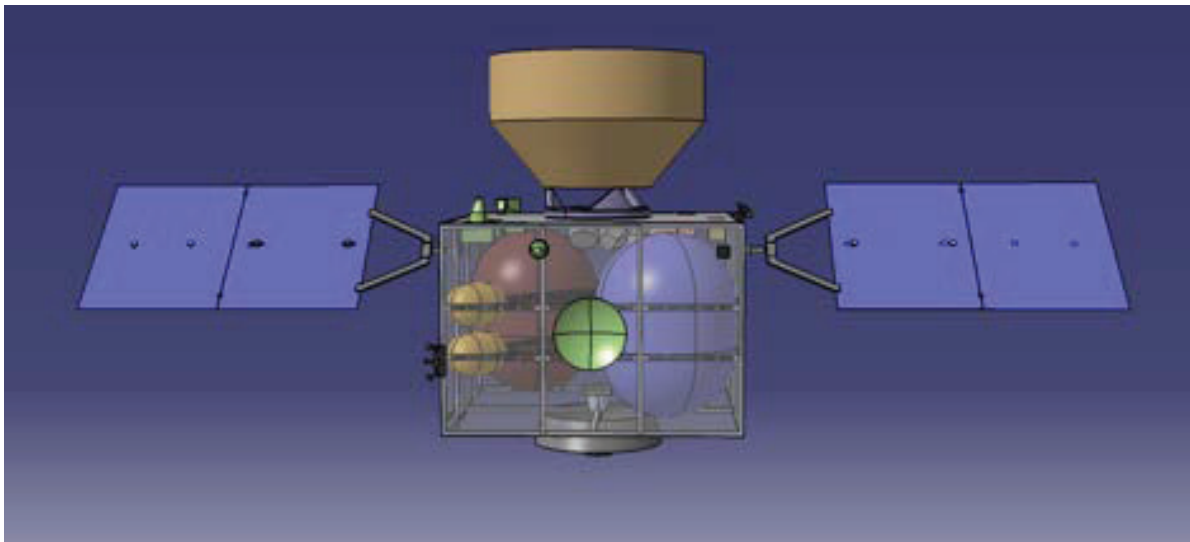


Fig. 4. Final Configuration of the PRIMAVERA Spacecraft (Face C)

For the first question, there were two possible configurations outlined, as seen in Figure 41. Configuration 1A offered as advantages a balanced weight distribution, ease in manufacturing and structural design (no lateral loads), and heritage in that it is a common arrangement of payload. The main disadvantage was the main attachment point between the spacecraft and the LVA was on a delicate surface (the location of the main nozzle). Configuration 1B solved the interface between the launcher and the spacecraft, by locating the attachment on a cleaner face. The main drawbacks of this configuration were the asymmetry in weight and the possibility of interference with the fairing.

Following the analysis, Configuration 1A was selected.

For the second question, in this case, two possible configurations were also studied, as seen in Figure 42. Configuration 2A comprised the jettisoning of the main propulsion system, with the advantage of a large weight reduction (and hence, an extended life and more effective use of the available resources). The main drawbacks of this case relied on the unfamiliarity of the process, the loss of one useful face of the spacecraft, and the changes of CoM positioning that would necessitate jettisoning ~250 kilograms in order to restore inertial stability. On the other hand, Configuration 2B allowed for a simpler propulsion subsystem, with shared tanks between secondary AOCs thrusters and the main thruster, as well as a risk reduction (less operations), at the cost of having to carry the extra dead weight.

Following the analysis, Configuration 2B was selected.

10.4 Final Baseline Design

The final baseline design of the PRIMAVERA spacecraft is shown in Figure 4, and Tables 29 and 30 — The first table corresponding to the equipment tray and the second to the exterior of the spacecraft. For illustrative purposes, Figures 50, 51 and 52 show three renders of the spacecraft in operation around Venus, plus the probe detachment.

10.5 Open Points

Within the scope of this study, a few open points remain for future iterations:

- The choice of the Payload Adapter PLA6 1194 is a conservative one - if the CoM criteria in the fairing were demonstrated to comply with another solution, some weight could be saved.
- The thermal subsystem modelling is not detailed enough.
- A revision on the alignment of the thrust vector and the CoM should be performed.
- The placement and attachment of the RCS thrusters should be studied in more detail.
- A study should be carried out on the efficiency of fuel storage for an improved internal configuration.

- A more detailed scientific payload could be added to the orbiter, with further relevant design work carried out here.

11. Cost Engineering

PRIMAVERA was initially defined as an M Class Mission (overall cost < 500 M€). However, its objectives were significantly more ambitious than those typically associated with missions in this class. Therefore, the challenge for the cost estimation team was to ensure that all the characteristics and requirements of the mission were included within that budget. Through an iterative cost engineering process and close collaboration with all relevant subsystems, parametric models (USCM8 and Quick-Cost Analysis) were first employed to subsequently carry out a more detailed bottom-up analysis. This comprehensive approach took into account all satellite subsystems, project phases, and the planning model philosophy, yielding the results shown in Figures 43 and 44, which break down the cost of spacecraft hardware and software and that of all the system level activities.

Following these two analyses, a prospective study was conducted to determine which launcher would be used for the mission. Both European and other regions' launchers were evaluated, but ultimately the decision was made to use Ariane 64 due to the size of the mission and the geographical return conditions imposed by ESA. This launcher has a total cost per launch of 120 M€. Operating costs were also calculated taking into account a transfer duration of 15.7 months, an operations phase of 7.5 months, and a combined science operations and post-operations phase of 12 months. This resulted in a budget of 80.1 M€ for the mission's operations.

Finally, a risk assessment was carried out using a three-point estimate with pessimistic and optimistic values for each of the above budgets. Thus, after reaching an agreement with the systems engineering team on the level of confidence required to obtain, the total budget was determined as shown in Figure 45. As a result, a total cost of 465.3 M€ was established, which already accounts for a 20% margin, providing a 70% confidence level. Consequently, PRIMAVERA can be classified as a Class M+ mission. While both its mass and mission concept align with Class L criteria, the project's cost falls at the upper threshold of Class M [13].

12. Systems Engineering

12.1 CONOPS

The CONOPS for the PRIMAVERA mission is a preliminary model used to assess worst case scenarios within

reasonable operational expectations. In this case, the main considerations are communications windows with the probe, and with Earth. The active component of the CONOPS is the satellite/orbiter, and the other parts of the system are the ground station and the probe. The modes of operations are as follows:

- LEOP - critical post-launch phase where operators take control of the satellite, activate systems, and guide it to its final orbit.
- IDLE - waiting for a communication window, while in nadir orientation.
- SAFE - protective state where a spacecraft shuts down non-essential systems and focuses on maintaining power, thermal balance, and communication after detecting a fault or anomaly.
- MAN - engage AOCS to move between nadir-facing and Earth-facing orientations.
- COM_PL - data transmission from probe to satellite.
- COM_DTE - transmit data to ground station on Earth.
- ECL - mode to handle situations arising from eclipse events.
- OFF - satellite decommissioning.

These modes are spread throughout the following mission phases (Figure 46):

- LEOP - LEOP and SAFE
- Platform (Orbiter) Commissioning - IDLE and SAFE
- Transfer to Venus - MAN and SAFE
- Payload Deployment - MAN, IDLE, and SAFE
- Venus Arrival - MAN and SAFE
- Transfer to Operational Orbit - MAN and SAFE
- Nominal Operations - COM_PL, COM_DTE, ECL, IDLE, and SAFE
- EOL - OFF

12.2 Mass & Power Budgets

With the payload mass of 1000 kg and the platform mass of 1150 kg, using a 20% system margin, the total dry mass at launch is 2494 kg. With the added mass and contingency from pressurant (20 kg), propellant (3754 kg), and the payload launch adapter (85 kg), a final wet mass of 6353 kg is obtained. A more detailed breakdown of the mass of each subsystem is available in Table 47. This puts the PRIMAVERA mission under the A64 Performance of 6900 kg with a launch excess margin of 8%. The power budget for the mission is similarly depicted in Table 48.

12.3 Design Process & COMET

Throughout the five day workshop, the team undertook a rigorous and iterative design process, refining the spacecraft's design through three major iterations. This iterative progress can be exemplarily seen in the evolving mass budget, illustrated in Figure 49, which highlights the improvements made across each iteration, reflecting the increasing accuracy and optimisation of the design.

To streamline the design process and facilitate seamless collaboration across multiple subsystems, ESA's Concurrent Model-based Engineering Tool (COMET) was employed. Developed specifically for use at ESA's Concurrent Design Facility (CDF), COMET plays a crucial role in enabling real-time, simultaneous access to up-to-date design information across all teams. This tool ensures that every subsystem has the latest data available, facilitating informed decision-making and reducing the potential for errors or outdated assumptions during the design process.

The design sprints were conducted under the supervision of a systems engineering team composed of engineers from ESA's CDF, who brought their expertise to the table and led key discussions. Their leadership was instrumental in guiding the team through critical design trade-offs and balancing the diverse requirements and constraints of the spacecraft. Through collaborative teamwork and the effective use of COMET, the team was able to navigate complex engineering challenges and arrive at an optimised design solution.

13. Conclusions

The PRIMAVERA study has demonstrated the effectiveness of concurrent engineering in rapidly developing a complex mission concept within a greatly constrained time frame (less than a week in this case). By working in parallel and continuously iterating across subsystems, the team was able to identify interdependencies early, manage design trade-offs efficiently, and converge on

a coherent architecture that balanced mass, power, and communications requirements. The exercise highlighted the importance of close coordination between propulsion, AOCS, and communications, as well as the value of digital tools such as COMET for tracking evolving parameters in real time. All of PRIMAVERA's contributors have gone on to employ concurrent engineering principles in their daily work, be that in industry or at university, either personally or in wider group-based projects. Moreover, many still encourage discussion on its pitfalls as well as its benefits, where they themselves have faced setbacks within the highly variable and intense environment that concurrent engineering can create. Given the teams' desire to seek improvements and innovations to current industry practices, it is hoped that PRIMAVERA can encourage further investigation into how large scale engineering design work may be conducted via novel methods within short time frames.

Looking ahead, the PRIMAVERA design offers a strong foundation for future Venus atmospheric mission proposals. The integration of a variable-altitude balloon probe with an orbiter provides a novel approach to in situ cloud exploration and could serve as a pathfinder for more ambitious long-duration platforms. With further refinement of propulsive efficiency, thermal resilience, and extended communications strategies, this mission architecture could evolve into a scalable framework for future atmospheric science at Venus and potentially other planetary bodies with dense atmospheres. Keeping longer-term scientific and exploratory endeavours in mind, the PRIMAVERA team hopes that this work may inspire future missions to both Venus and beyond, revitalising humankind's desire to venture across the final frontier.

Acknowledgements

The entire PRIMAVERA team expresses its sincere gratitude to the European Space Agency — particularly the Education Department — for enabling the organisation and successful execution of the Concurrent Engineering Workshop 2024, in addition to providing individual sponsorship for each PRIMAVERA team member to attend the event. Appreciation is also extended to the staff at ESEC-Galaxia and ESEC-Redu for their generous hospitality, and to Mr Alex Rosenbaum, Ms Flavie Rometsch, Mr Salvatore Vivenzio, Mr Michel Van Pelt, Ms Sofia Santos, Mr Dario Farina and Ms Alessandra Marzolini for their outstanding stewardship, knowledge transfer, and expert guidance throughout the workshop. Furthermore, the team is deeply grateful to the International Astronautical Federation and the International Astronautical Congress for providing the

opportunity for PRIMAVERA to engage with a global platform that fosters collaboration, innovation, and knowledge exchange within the space community. IAC 2025 represents the first major international conference experience for many members of the team — a milestone for which they are both honoured and profoundly thankful.

Mr M. Nicoll would like to extend his personal thanks to the Institute of Mechanical Engineers (IMechE) for aiding in the facilitation of his attendance at IAC 2025.

Mr S. Leader would similarly like to thank the Canadian Space Agency (CSA) for their contributions towards his attendance at IAC 2025, as well as to the 2024 iteration of ESA's Concurrent Engineering Workshop itself.

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Mr N. Thipaharan would like to thank Professor William E. Crofts for enabling access to both the ESA workshop, and the IAC 2025.

Appendix A – Introduction & Concept

ID	Requirement
MIS-R-010	The mission shall be able to carry and deliver the floating platform into the Venusian atmosphere at 250 km altitude above the Venusian equator.
MIS-R-020	The mission shall be launched before 2035 .
MIS-R-030	The mission shall provide at least 1 Venusian year of science operations at Venus.
MIS-R-040	The mission shall be able to relay all data from the floating platform to Earth.
MIS-R-050	The mission shall be able to provide all payload data from the floating platform to Earth within 7 days.
MIS-R-060	The mission shall be able to provide additional remote science from the mothership, with a TBD payload.
MIS-R-070	The mission shall be launched with a European launcher .
MIS-R-080	The mission shall make use of European ground stations (ESTRACK network).
MIS-R-090	The mission shall adhere to Planetary Protection requirements for Venus.

Table 1. Mission Requirements

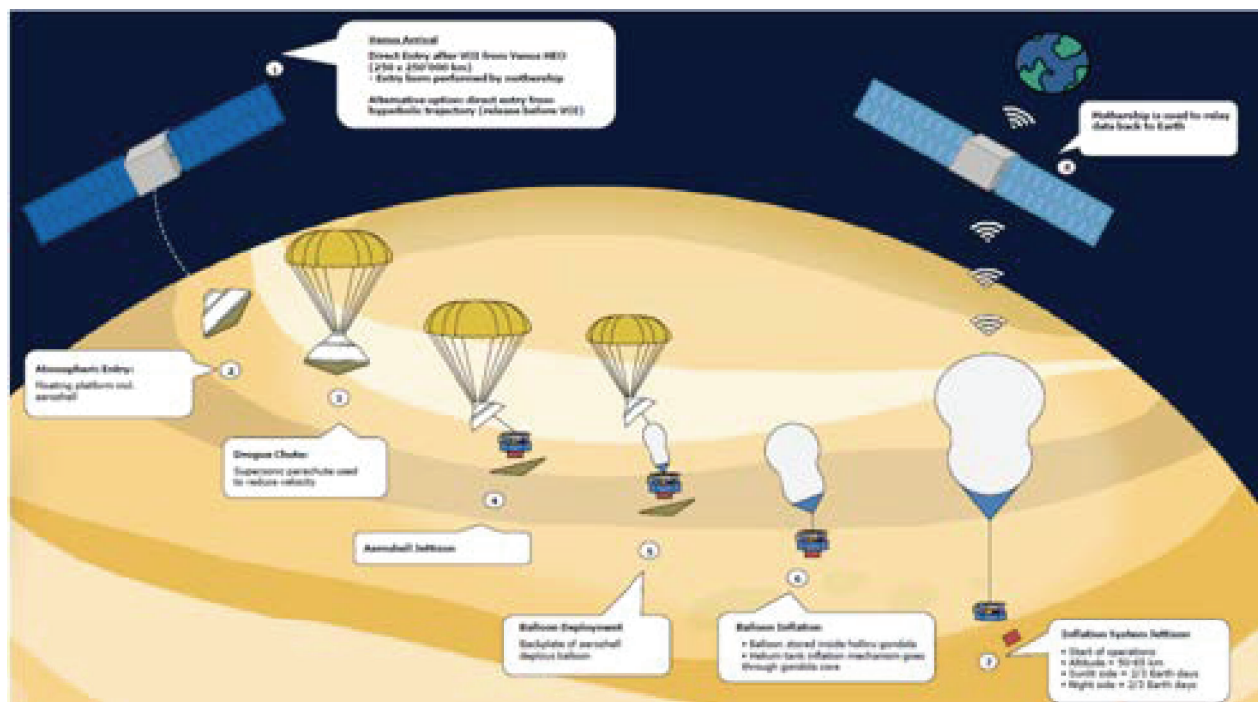


Fig. 5. Preliminary CONOPS

Specification	Value/Description
Mass	1000 kg total (floating platform + aeroshell)
Dimensions & Configuration	(See Figure 6)
Mechanical Interface to Orbiter	Spin & eject mechanism (assume ca. 30 kg) - part of mothership design.
Electrical Interface to Orbiter	TBD
Power Required during Transfer	100 W (constant)
Data Generated during Operations (e.g. per hour, day)	Data rate: 100 kbps
Communications to Orbiter	S-band (others may be considered if desired).
Expected Operational Altitude	Balloon system to operate between 50-65 km (where conditions are alike Earth sea-level). The altitude can be controlled.
Expected Operational Longitude/Latitude	The floating platform is expected to make one “tour” (i.e. rotation) of longitudes of the planet every 4-6 days - this is due to strong winds. These are not controllable.
Injection/Release Conditions from Carrier	Release after Venus orbital injection at 250 km. For release, carrier must be on assumed minimum 250 * 250,000 km orbit.
Payload Mass Allowed	55 kg
Payload Power Allowed	50 W day, 20 W night (both continuous)
Operational Lifetime	Minimum 1 Venusian year (225 Earth days)

Table 2. Floating Platform Specifications

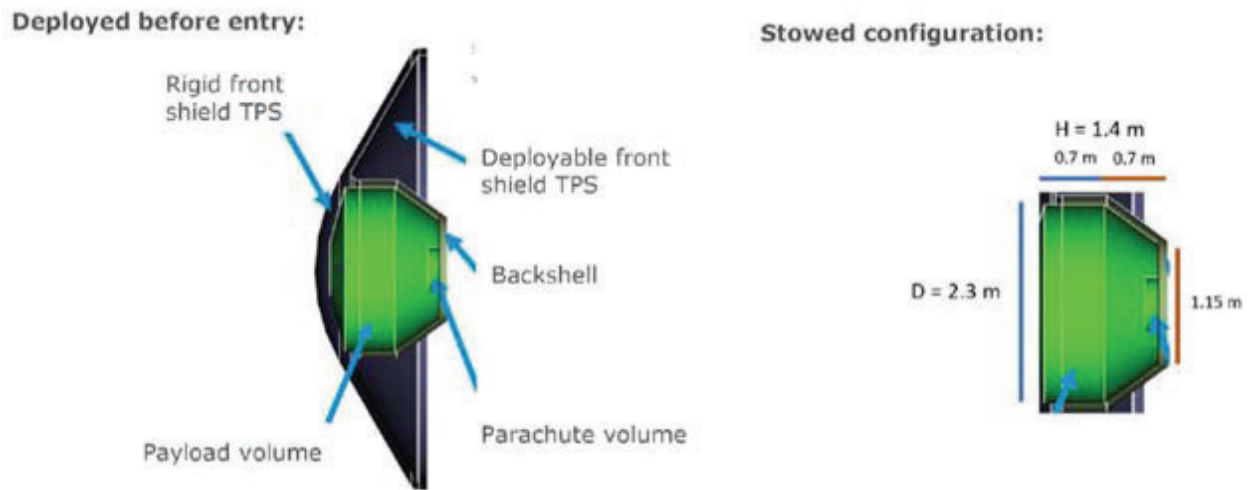


Fig. 6. Payload Configuration

Appendix B - Trajectory Analysis

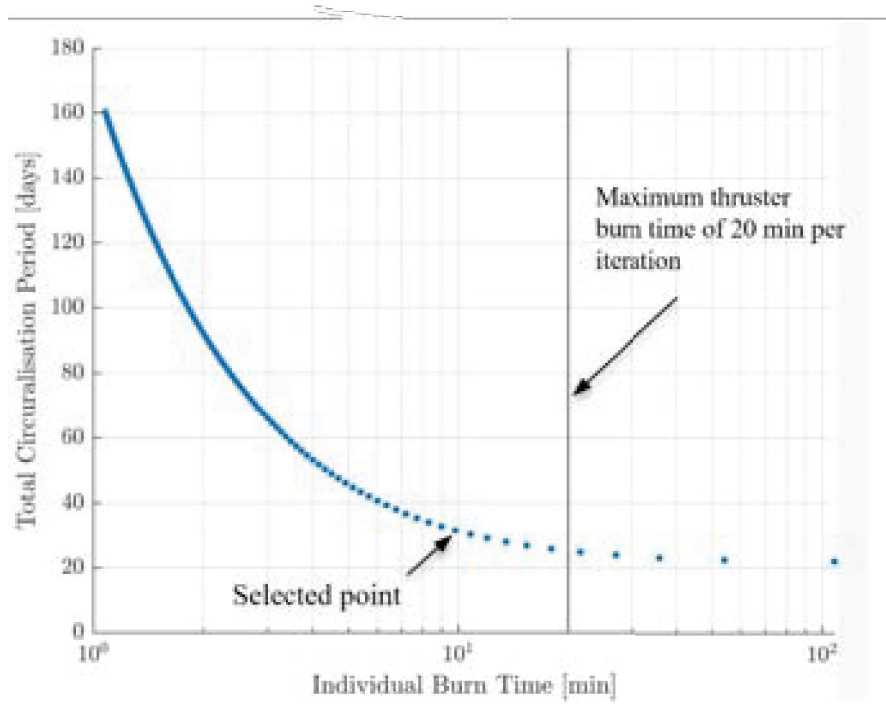


Fig. 7. Manoeuvre Time

Appendix C - Propulsion

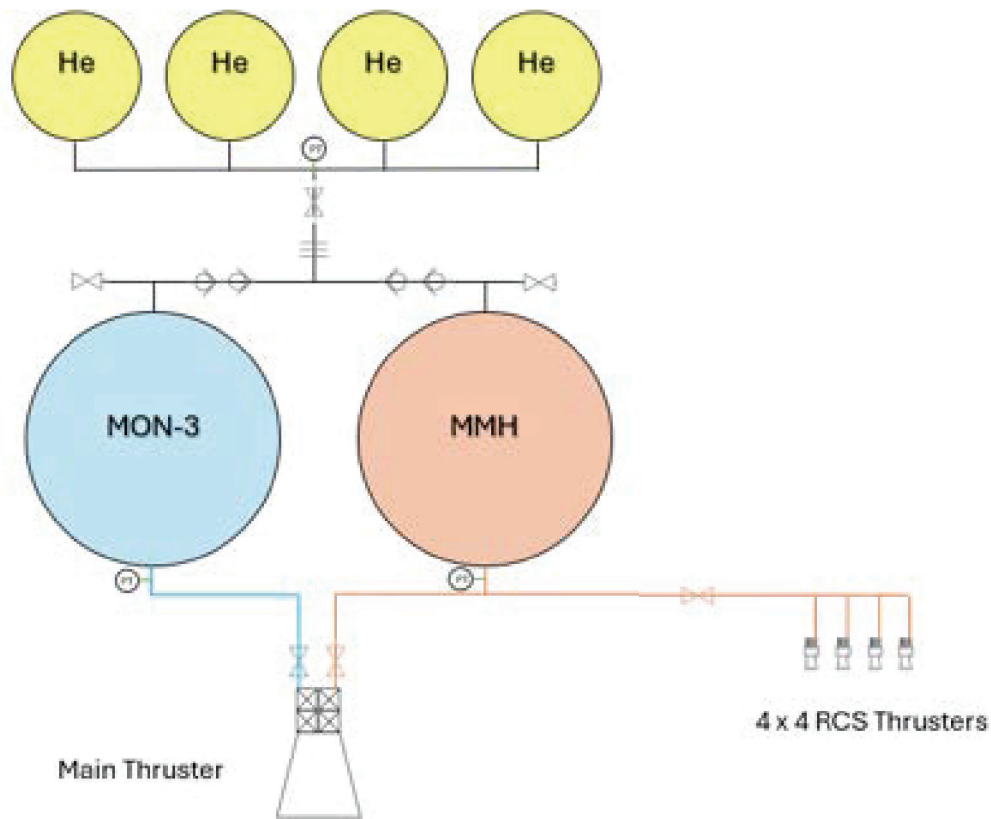


Fig. 8. Propulsive Apparatus



Fig. 9. LEROS 4 Thruster [5]

Appendix D - Power

Mission Phase	Total Pwr Req. (W)
LEOP	337
SAFE	138
COMMS_DTE	527
ECL	4665
COMMS_PL	437
IDLE	519
MANOEUVRE	479

Table 3. Peak Power Requirements by Mission Mode

Appendix E - Communications & Data Handling

ID	Description
COMM-010	The orbiter shall relay all data from the Payload to Earth, generated at 100 kbps.
COMM-020	The orbiter shall provide all Payload data to Earth within 7 days of generation.
COMM-040	The probe can store the data generated in each cycle to transmit to the orbiter.

Table 4. Comms Design Drivers

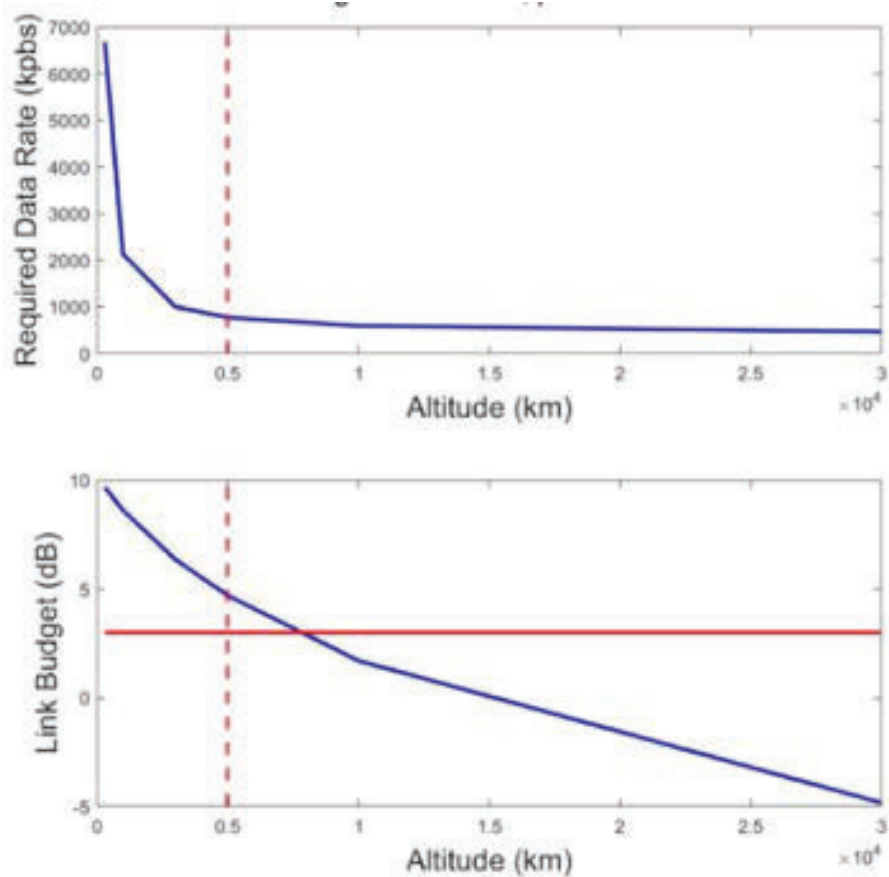


Fig. 10. Comms Trade-Off Analysis

Appendix F - Attitude & Orbital Control Subsystems

ID	Description
AOCS-DD-001	Frequent slewing to provide communication access for the high-gain antenna (HGA) for weather balloon uplink/Earth downlink. Slew frequency set by COM bit-rate.
AOCS-DD-002	Slewing operations require reaction wheel off-loadings in at worst case, once per day, with as little parasitic ΔV generated as possible (ideally < 1 mm/s/day). This directly impacts the sizing and selection of the reaction control system (RCS) thrusters.

Table 5. AOCS Design Drivers

ID	Description
AOCS-DA-001	For assessment of environmental disturbances and torque requirements, the spacecraft inertia matrix was assumed to be the following: [6840 0 0; 0 8370 0; 0 0 10570] kg m ² .
AOCS-DA-002	To assist in the determination of control requirements, magnetospheric and atmospheric disturbances were deemed negligible.
AOCS-DA-003	The maximum spacecraft cross-section was assumed to have an area of 21.7 m ² .
AOCS-DA-004	The spacecraft average reflectance factor was assumed to be 0.58.

Table 6. AOCS Design Assumptions

ID	Description
AOCS-REQ-010	The AOCS subsystem shall provide hardware and associated onboard software to acquire, control and measure the required spacecraft attitude during all phases of the mission, and to control and monitor all the necessary ΔV manoeuvres for the complete mission (including potential extended lifetime).
AOCS-REQ-020	In Venusian orbit, the PRIMAVERA spacecraft shall be 3-axis stabilised with the medium-gain antenna (MGA) pointing towards nadir, and solar arrays to celestial poles.
AOCS-REQ-030	In Venusian orbit, the wheel off-loading shall not take place more often than once per slew, and the parasitic ΔV generated by the RCS thruster apparatus during wheel off-loading shall be limited to 1 mm/s/day.
AOCS-REQ-040	During communication windows, the contribution of the AOCS to the APE of the HGA shall not exceed 30 arcsec half cone.

Table 7. AOCS Subsystem Requirements

Steps	Inputs	Outputs
1a) Define control modes. 1b) Define or derive system-level requirements by control mode.	Mission requirements, mission profile, type of insertion for launch vehicle.	List of different control modes during mission. Requirements and constraints.
2) Quantify disturbance environment.	Spacecraft geometry, orbit, solar/magnetic models, mission profile.	Values for torques from external and internal sources.
3) Select type of spacecraft control by attitude control mode.	• Payload, thermal and power needs. • Orbit, pointing direction. • Disturbance environment. • Accuracy requirements.	Method for stabilisation and control: three-axis, spinning, gravity gradient, etc.
4) Select and size AOCS hardware.	Spacecraft geometry and mass properties, required accuracy, orbit geometry, mission lifetime, space environment, pointing direction, slew rates.	• Sensor suite: Earth, Sun, inertial, or other sensing devices. • Control actuators: reaction wheels, thrusters, magnetic torquers, etc. • Data processing avionics, if any, or processing requirements for other subsystems or ground computers.
5) Define determination and control algorithms (not strictly applicable within condensed, high-level approach adopted for the development of PRIMAVERA).	Performance considerations (stabilisation method(s), attitude knowledge and control accuracy, slew rates) balanced against system-level limitations (power and thermal needs, lifetime, jitter sensitivity, spacecraft processor capability).	Algorithms and parameters for each determination and control mode, and logic for changing from one mode to another.
6) Iterate and document.	All of the above.	• Refined mission and subsystem requirements. • More detailed AOCS design. • Subsystem and component specifications.

Table 8. AOCS Iteration Process [14]

Mode	Description
Orbit Insertion	Period during and after boost while the spacecraft is brought to final orbit. Options include no spacecraft control, simple spin stabilisation of solid rocket motor, and full spacecraft control using liquid propulsion system. May drive certain aspects of AOCS design.
Acquisition	Initial determination of attitude and stabilisation of vehicle for communication with ground and power generation. Also may be used to recover from power upsets or emergencies.
Normal Mission, On-Station	Used for the vast majority of the mission. Requirements for this mode should drive system design.
Slew	Reorienting the vehicle when required.
Contingency/Safe	Used in emergencies if regular mode fails or is disabled. Will generally use less power or fewer components to meet minimal power and thermal needs.
Special	Requirements may be different for special targets or time periods, such as when the satellite passes through a celestial body's shadow, or umbra.

Table 9. AOCS Control Modes

Torque	Value
Gravity Gradient Torque (Nm)	7.929×10^{-4}
Solar Radiation Pressure Torque (Nm)	1.089×10^{-3}
Total (RSS; Nm)	1.350×10^{-3}
Slew Torque (Nm)	8.201×10^{-2}

Table 10. Torque Data

Parameter	Value
Integrated Angular Momentum (Nms)	4.14×10^{-1}
Momentum Cyclical Component (Nms)	3.32×10^{-1}
Momentum Secular Component (Nms)	8.29×10^{-2}
Max. Torque in an Orbit (Nm)	1.95×10^{-3}

Table 11. Torque & Momentum Data in an Orbit (Integration)

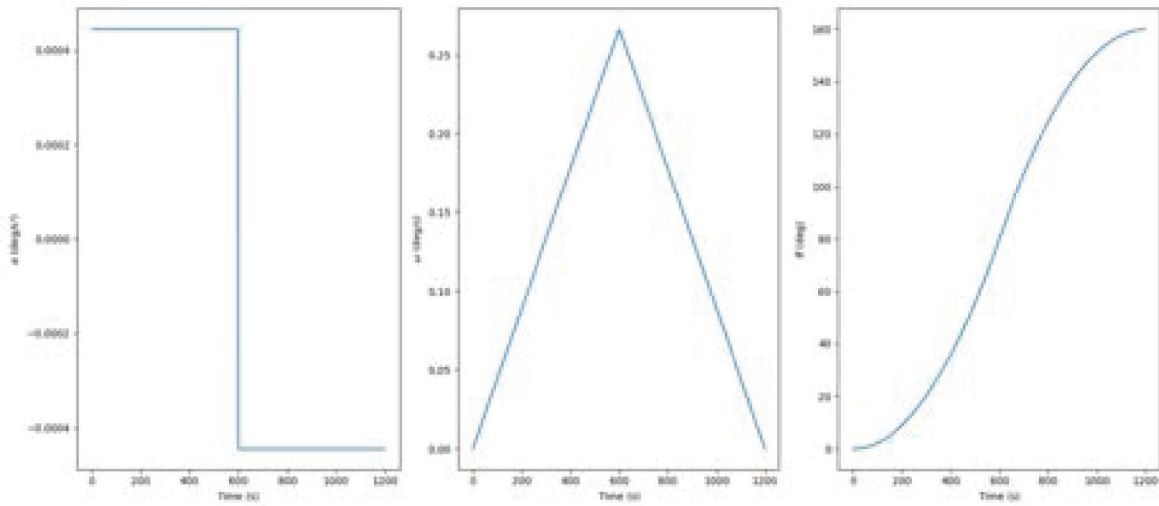


Fig. 11. 'Bang-Bang' Slew Profile



Fig. 12. Collins Aerospace RSI 68-75/60 Reaction Wheel [15]

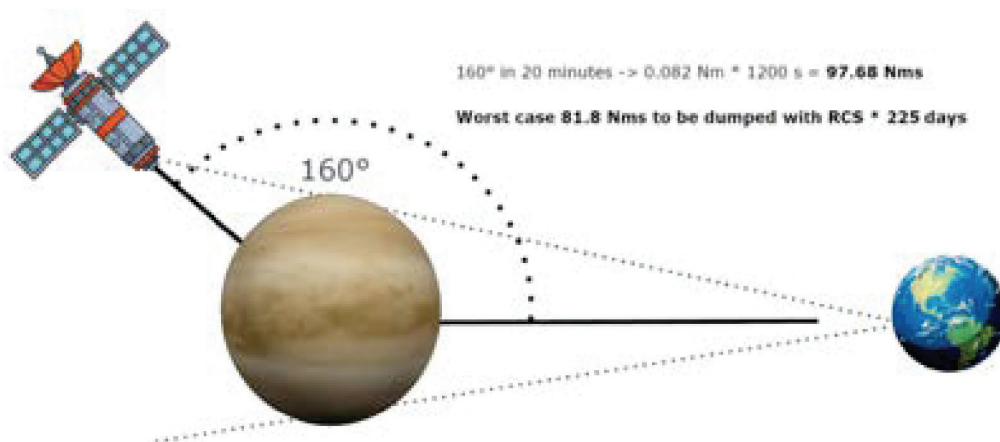


Fig. 13. Worst Case Slew Profile

Parameter	Value/Description
Item	RSI 68-75/60
Manufacturer	Collins Aerospace
Angular Momentum (Nms)	68
Operational Speed Range (RPM)	±6000
Quantity	4 (Tetrahedral Configuration)
Unit Mass (kg)	8.5
Dimensions (m)	0.347 (D) x 0.118 (H)
Power (Nominal/Peak, W)	20/90
Redundancy	3 + 1 (active)

Table 12. Reaction Wheel Data [16]

Parameter	Value/Description
Item	MRE-1.0 Monopropellant Thruster
Manufacturer	Northrop Grumman
Thrust at Max. Op. Pressure (N)	4.5
Burn Time (s)	3
Propellant	Hydrazine
Steady State ISP (s)	220
Op. Pressure Range (psia)	50-400
Max. Throughput (kg)	122
Max. Op. Cycles	421000
Power (W)	8.5
Unit Mass (kg)	1
Envelope (W x L, mm)	190 x 38
Total RCS Prop. Mass (kg)	10
Thruster Configuration	Quad
Thruster Misalignment (deg)	90*
Redundancy	2 + 2 (passive)

Table 13. RCS Hardware Data [17]

*(Alteration required to be requested from COTS component. Typically provided as parallel pairs.)



Fig. 14. ST200 Star Tracker [18]

Parameter	Value/Description
Item	ST200 Star Tracker
Manufacturer	AAC Clyde Space
Quantity	2
Unit Mass (kg)	0.042
Dimensions (mm)	29 x 29 x 38.1
Power (Nominal/Peak, W)	0.67/1
Redundancy	1 + 1 (passive)

Table 14. ST200 Star Tracker Data [19]



Fig. 15. Bradford Space CSS Unit [20]

Parameter	Value/Description
Item	Coarse Sun Sensor
Manufacturer	Bradford Space
Quantity	2
Unit Mass (kg)	0.215
Dimensions (mm)	110 x 110 x 30
Power (Nominal/Peak, W)	Nil. (passive, analog unit)
Redundancy	1 + 1 (passive)

Table 15. Bradford Space CSS Data [21]



Fig. 16. ASTRIX 1000 Gyroscope [22]



Fig. 17. AI-Q-2030 Accelerometer [23]

Parameter	Value/Description
Item	ASTRIX 1000
Manufacturer	Airbus Defence and Space
Quantity	2
Unit Mass (kg)	4.5
Dimensions (mm)	263 x 192
Power (W)	13.5
Redundancy	1 + 1 (active)

Table 16. ASTRIX 1000 Gyroscope Data [24]

Parameter	Value/Description
Item	AI-Q-2030
Manufacturer	InnaLabs
Quantity	6
Unit Mass (kg)	0.071
Dimensions (mm)	25.45 x 14.85
Power (Nominal/Peak, W)	0.2/0.48
Redundancy	3 + 3 (active)

Table 17. AI-Q-2030 Accelerometer Data [25]

Appendix G - Ground Segment

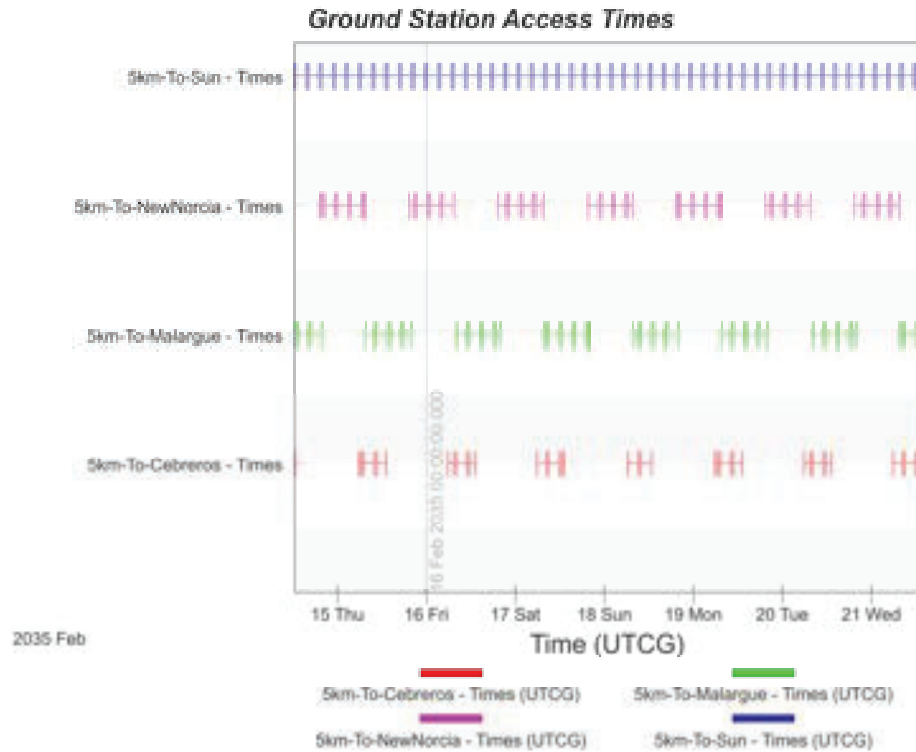


Fig. 18. Access Times



Fig. 19. ESA ESTRACK Station New Norcia, Australia [26]



Fig. 20. ESA ESTRACK Station; Cebreros, Spain [27]

Appendix H - Structures & Mechanisms

ID	Requirement
STR-010	The spacecraft primary structure stiffness shall comply with A6.4 launch vehicle stiffness requirement.
STR-020	The spacecraft shall fit in the launcher's fairing in the stowed configuration (cf. Annexe).
STR-030	The spacecraft CoG position in height and its lateral position in the LV's coordinate system shall comply with limits defined (cf. Annexe).
STR-040	The spacecraft adapter shall comply with a standard A6.4 LVA.

Table 18. Mission Structural Requirements

Structure	Number
Skin	6
Stringers	12
Rings, Frames & Fasteners (RFF)	1
Propulsion Structure	2
Nozzle Structure	1
Payload Adapter	1

Table 19. Additional Structures Information

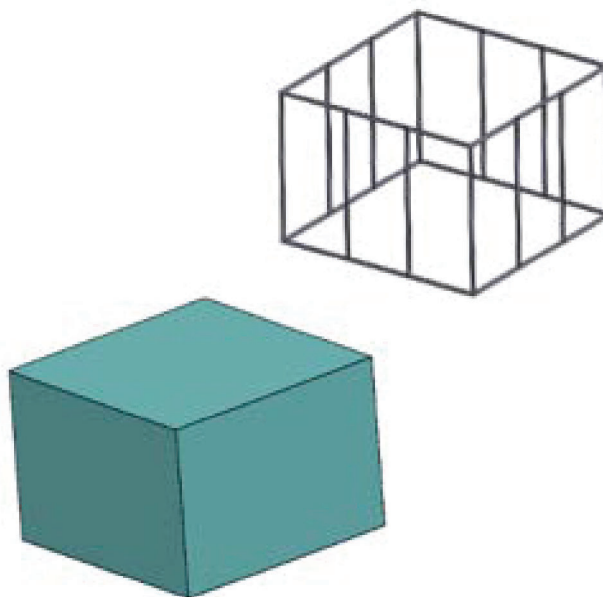


Fig. 21. Satellite Structure: Skeleton & Skin

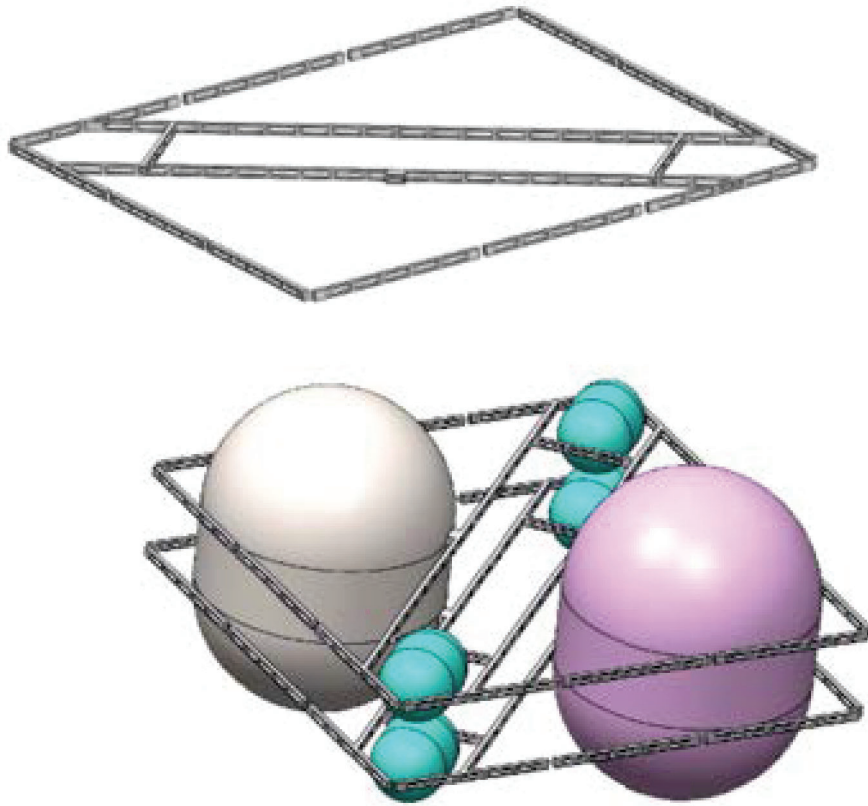


Fig. 22. Propulsion System Support Structure Configuration

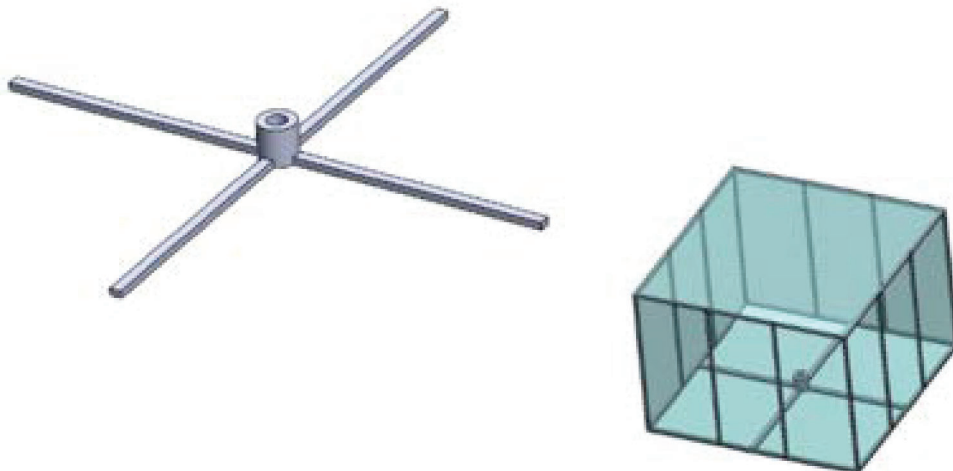


Fig. 23. Thrust Structure

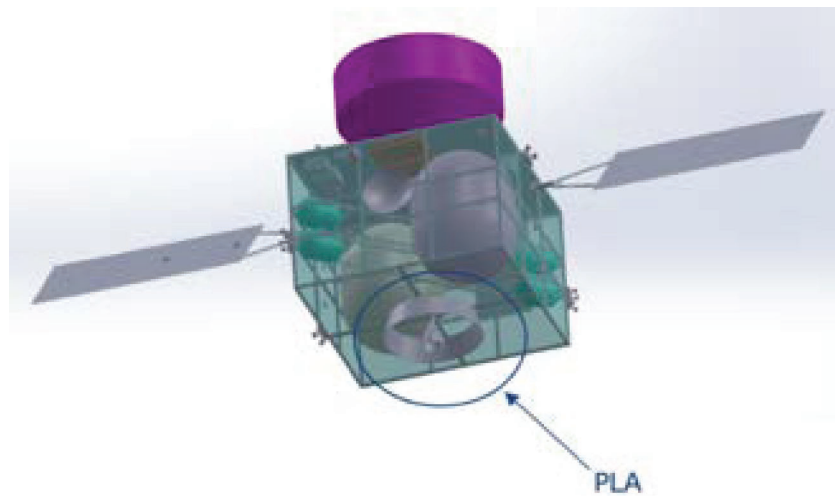


Fig. 24. Payload Adapter

ID	Requirement
MEC-010	All mechanisms shall be deployed in LEOP.
MEC-020	The orbiter antenna shall follow the floating platform across 120 deg. angle in COMS P/L.
MEC-030	The spacecraft shall include a spin and eject mechanism for the Floating Platform.
MEC-040	The mechanism for the Floating Payload shall have a mass of 30 kg.
MEC-050	The HGA shall have a fixed position.
MEC-060	Solar arrays shall be able to rotate 350 degrees.
MEC-070	The Solar Array Deployment Mechanism shall deploy a solar array of 31.5 kg.
MEC-080	The SADM shall deploy a solar array of two panels.

Table 20. Mechanism Requirements



Fig. 25. Antenna Pointing [28]



Fig. 26. ARQUIMEA's REACT (HDRM) [29]



Fig. 27. 3/5 Solar Array Drive Assembly MOOG [30]

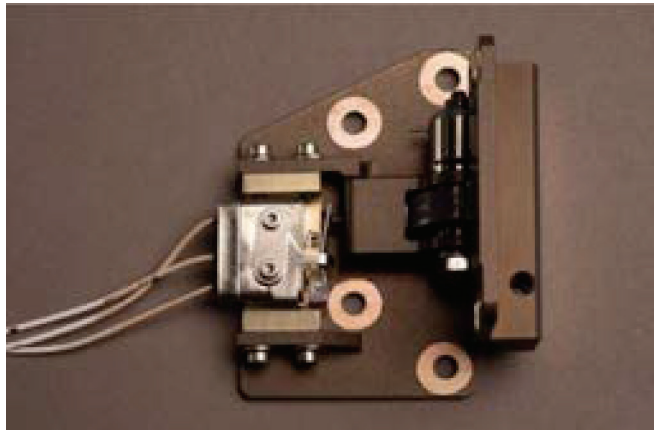


Fig. 28. ARQUIMEA's Hinge [31]

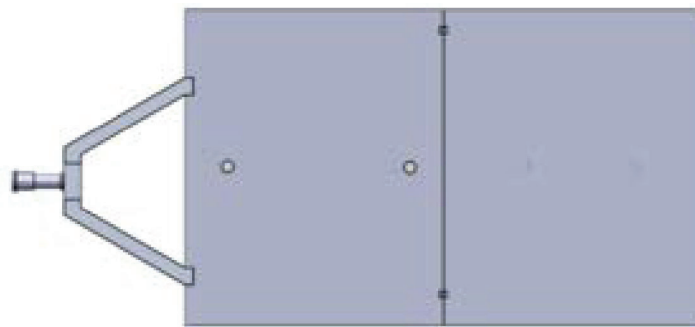


Fig. 29. Yoke & Solar Panels

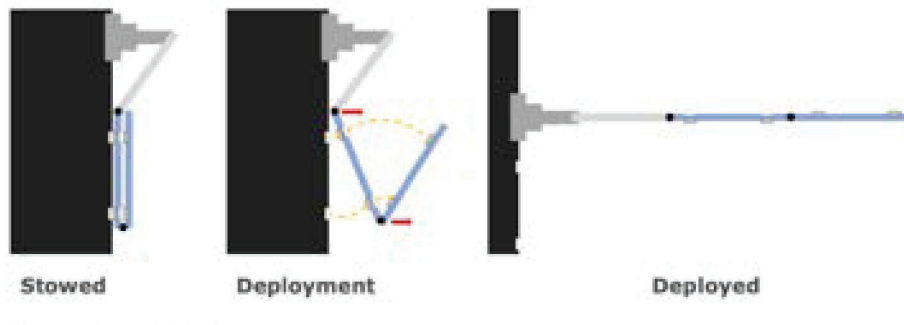


Fig. 30. Solar Panel Deployment

Mechanism	Quantity	TRL
Antenna Pointing	1	9
Floating Platform Deployment	1	7
HGA Bracket	1	8
Solar Array Deployment - HDRMs	8	9
Solar Array Deployment - Hinges	8	9
Solar Array Adjustment	2	9
Yoke	2	5

Table 21. Mechanisms Summary

Appendix I - Thermal Control

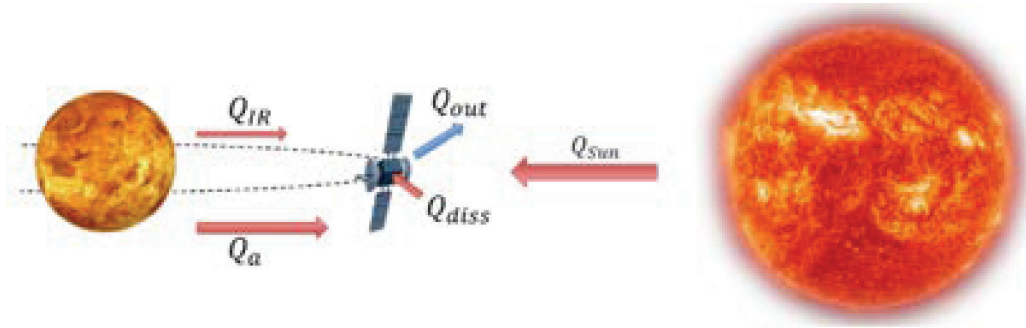


Fig. 31. Hottest Case Thermal Management

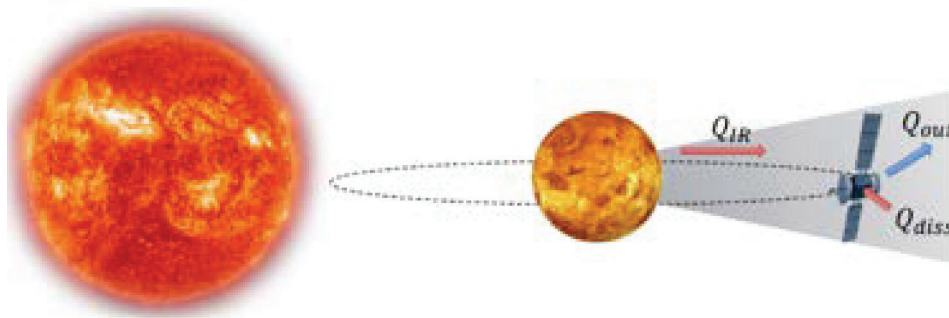


Fig. 32. Coldest Case Thermal Management

Design driver	Primary implications
Limit operating temperatures	$A_{\text{radiator}}, Q_{\text{heater}}$
Orbit altitude	Q_{IR}
Spacecraft orientation	$Q_{\text{Sun}}, Q_{\text{a}}, Q_{\text{IR}}, Q_{\text{out}}$
Spacecraft geometry	$Q_{\text{Sun}}, Q_{\text{a}}, Q_{\text{IR}}, Q_{\text{out}}$
Spacecraft mass	Q_{heater}
Spacecraft power consumption	$A_{\text{radiator}}, Q_{\text{heater}}$
Spacecraft insulation material	Q_{out}
Radiator material	A_{radiator}
Solar array geometry	$T_{\text{min,SA}}, T_{\text{max,SA}}, M_{\text{surface finisher}}$
Solar array efficiency	
Solar cells material	
Solar array surface finisher	

Fig. 33. Thermal Subsystem Design Drivers

Options definition:	1/2 Mil Kapton	1 Mil Kapton	2 Mil Kapton	3 Mil Kapton	5 Mil Kapton	Silvered Teflon Aluminized Teflon	Beta Cloth	Astro Quartz	Weight Factor	
Mass	10	9	8	6	3	2	3	5	1	0.50
Cost	5	4	3	2	1	2	2	5	6	0.36
Power (ECL)	10	8	7	7	6	3	4	5	4	0.07
Equilibrium temperature	1	2	2	3	4	9	8	8	7	0.07
Total score	8	7	6	4	3	3	3	5	3	
Ranking	1	2	3	5	8	8	7	4	6	

Fig. 34. Insulating Material Trade-off Analysis

	Material	T _{max}	T _{min}
Optical Solar Reflectors	8mil quartz mirr	14.25189	-120.892
	Quartz mirr	28.96877	-120.892
	2mil silv Tefl	20.27488	-123.581
	3mil silv Tefl	15.06722	-121.23
	2mil alu Tefl	33.87473	-123.581
White Paints	5mil alu Tefl	27.53011	-121.23
	813G-LO	45.35106	-120.102
	PCBZ	36.65731	-119.806
Black Paints	Z93	36.52596	-119.112
	ZOT	38.89804	-119.246
	Chemg. A276	47.88853	-119.682
Aluminized Kapton	Chemg. Z306	132.8881	-119.521
	3M Black Velvet	140.3088	-120.254
	1/2 mil alu Kapt	86.02764	-126.384
Metallic	1mil alu Kapt	84.79125	-123.541
	2 mil alu Kapt	84.54388	-121.769
	5mil alu Kapt	85.39734	-119.953
Miscellaneous	Vap-dep aluminum (VDA)	70.63483	-181.5
	Bare Alu	67.77741	-161.333
	Vapor-dep gold	100.9429	-187.398
	SiO ₂ on VDA t	79.39443	-137.202
	FSS-69	56.40209	-195.235
	1/4 mil aluminized Mylar	21.55872	-134.9
	Beta cloth	65.40371	-119.953
	Astro Quartz	51.40846	-120.892
	TiNOX	212.9953	-176.753
	Maxorb	200.6403	-161.333

Fig. 35. Solar Array Reflector Analysis

Trade-off table	Options definition:	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6	Weight Factor (1..10)
		8mil quartz mirr	Z93	Chemg. Z306	5mil alu Kapt	Bare Alu	Beta cloth	
Key trade-off Parameter	Mass	5	10	9	8	4	7	0.35
	Cost	2	6	7	4	10	8	0.05
								0.00
	Equilibrium temperature	6	10	9	8	3	7	0.60
								0.00
								0.00
								0.00
								0.00
								0.00
	Total score	5	10	9	8	4	7	1.00
	Ranking	5	1	2	3	6	4	

Fig. 36. Solar Array Reflector Trade-off

Parameter	Value
Total Surface Area (m ²)	41.43
Effective Density (kg/m ³)	1420
No. Layers	25
Total Mass (kg; 10% Margin)	22.1
α (Aluminised Kapton ½ Mil)	0.34
ϵ (Aluminised Kapton ½ Mil)	0.55
Maximum Temp. Achieved (°C)	35

Table 22. Main Body Insulation

Parameter	Value
Thermal Conductivity (W/mK)	0.024
Density (kg/m ³)	35
Thickness (cm)	2
Mass (kg)	5.8
Tank Equilibrium Temp. (°C)	-9.20

Table 23. Individual Component Insulation

Parameter	Value
Area (m ²)	0.87
Emissivity, ϵ	0.8
Absorptivity, α	0.09
Unit Mass (kg/m ²)	0.350
Mass (kg)	0.3045
Maximum Temp. Achieved (°C)	30

Table 24. Radiating Surfaces

Parameter	Value
Area (m ² ; given by SA)	200
Emissivity, ϵ	0.92
Absorptivity, α	0.2
Mass per Unit Area (kg/m ²)	110.16
Mass (kg)	0.138
SA Equilibrium Temperature Range (K)	-120 to 150

Table 25. Internal Paint Coverage

Parameter	Value
No. Items	TBD
Total Length (m)	11.75
Density (kg/m ³)	1355
Total Mass (kg)	111.38

Table 26. Heat Pipes

Parameter	Value
No. Items	TBD
Heat Power Required (kW; 5% Margin)	3.609
Total Mass (kg; 5% Margin)	24

Table 27. Heating Elements

Appendix J - Configuration

ID	Requirement
CONF-010	The probe shall fit in the selected launcher fairing.
CONF-020	The interface to the launcher shall be based on the available existing standard interfaces.
CONF-030	The configuration shall accommodate all spacecraft equipment to comply with mission objectives, as well as power and communication requirements.
CONF-040	The configuration shall provide an unobstructed FOV for the sensors and the antennas.
CONF-050	The configuration shall provide unobstructed deployment for the mechanisms.
CONF-060	The configuration shall provide an unobstructed position for the thruster sets (main and RCS) to fulfill the mission objectives.

Table 28. Mission Configuration Requirements

Power subsystem	AOCS subsystem	Communications and Data Handling subsystem
Batteries	Reaction wheels	On-board computer
PCDU	Gyroscopes	Transponders (S, X and K _a Bands)
-	Accelerometers	TWTA

Table 29. Elements in the Equipment Tray

Power subsystem	AOCS subsystem	Communications and Data Handling subsystem	Thermal subsystem
Solar arrays	Star trackers	LGA	Radiator
-	Sun sensors	MGA + mechanism	-
-	RCS thrusters	HGA	-

Table 30. Elements in the External Faces

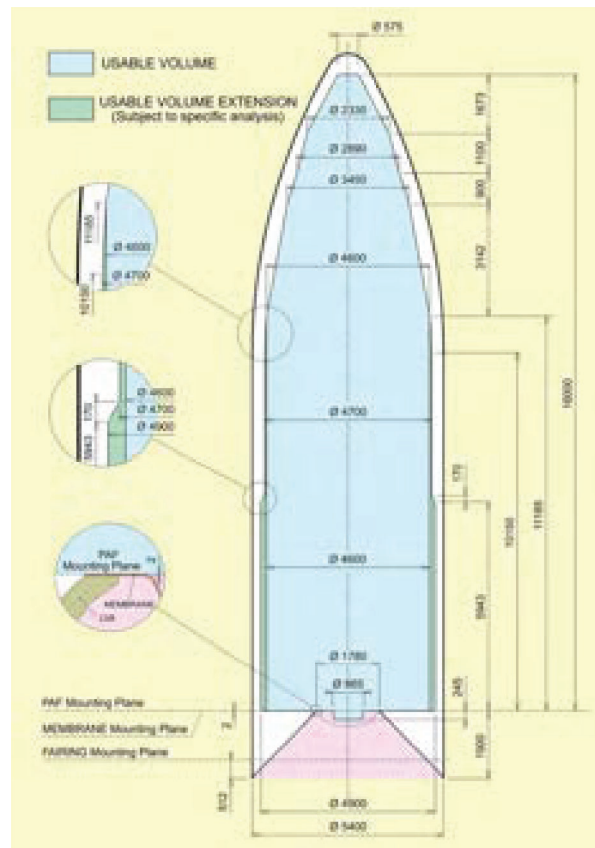


Fig. 37. Usable Volume within the Ariane 6 Long Fairing [32]

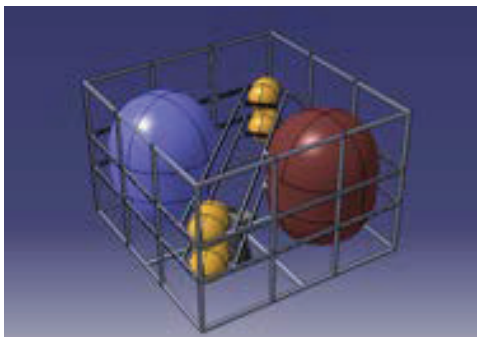


Fig. 38. Internal Distribution (Fuel Tanks)

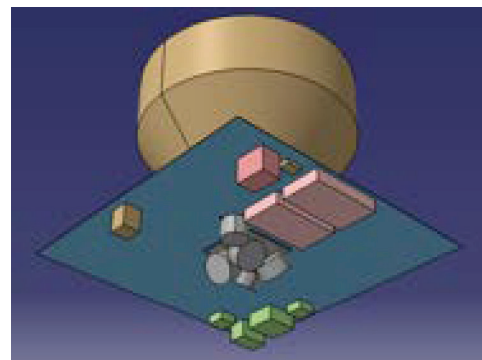


Fig. 39. Equipment Tray (Power Elements in Pink, AOCS in Gray, Communications in Green and OBC in Yellow)

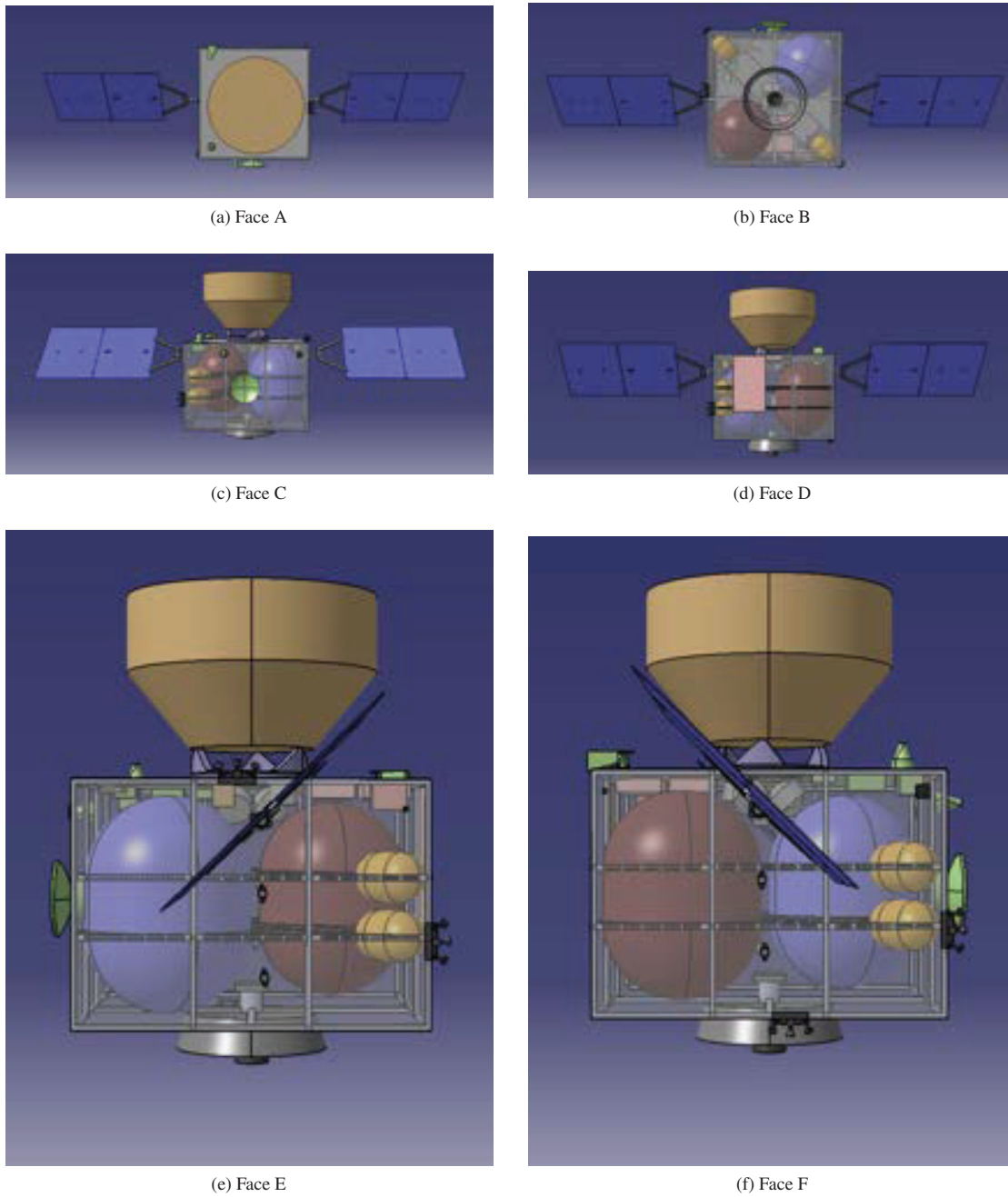
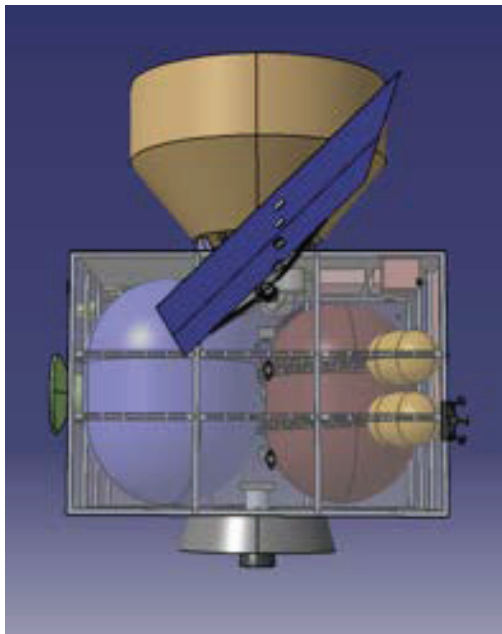
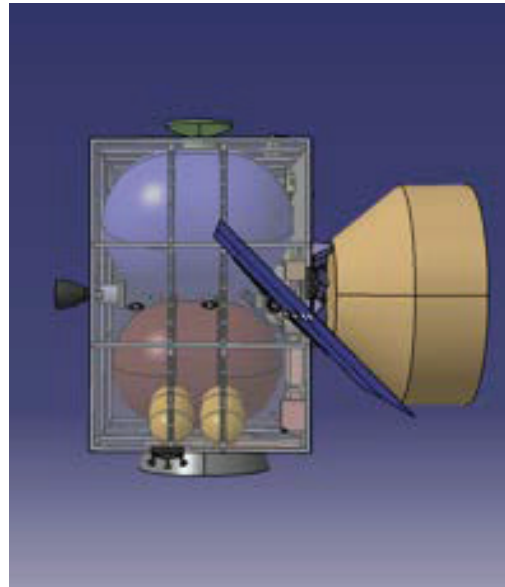


Fig. 40. Differing Views of the Spacecraft

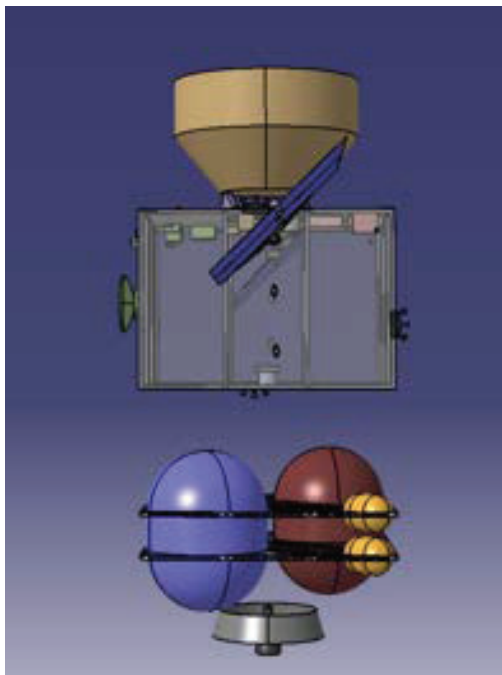


(a) Configuration 1A

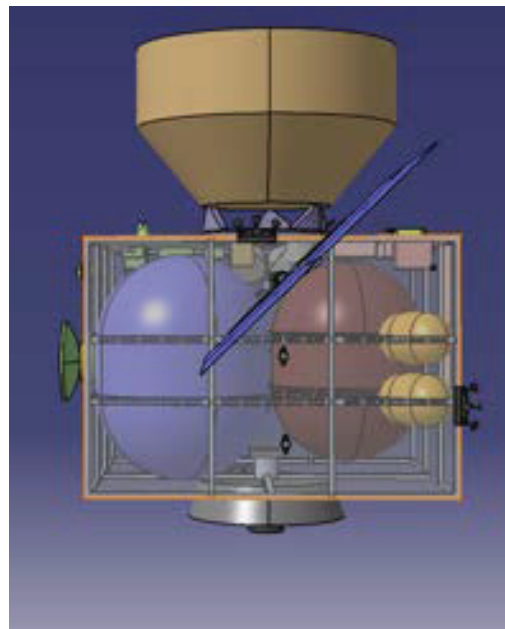


(b) Configuration 1B

Fig. 41. Configurations Compared in Trade-off 1



(a) Configuration 2A



(b) Configuration 2B

Fig. 42. Configurations Compared in Trade-off 2

Appendix K - Cost Engineering

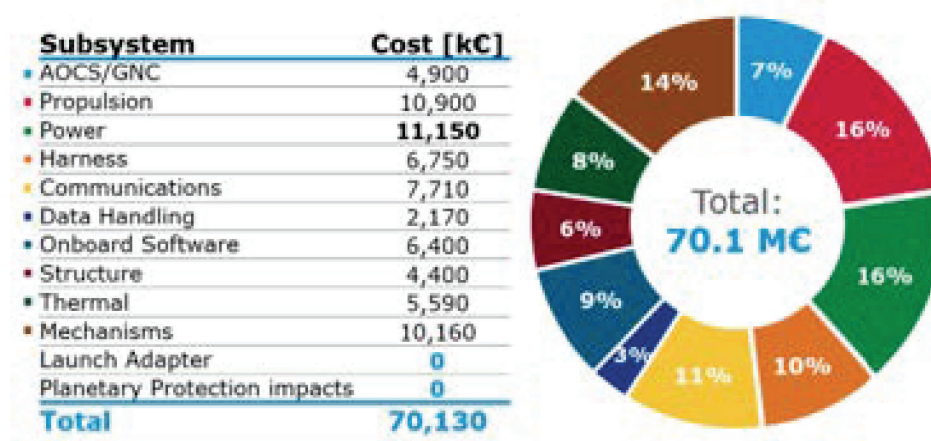


Fig. 43. Subsystems Cost Breakdown

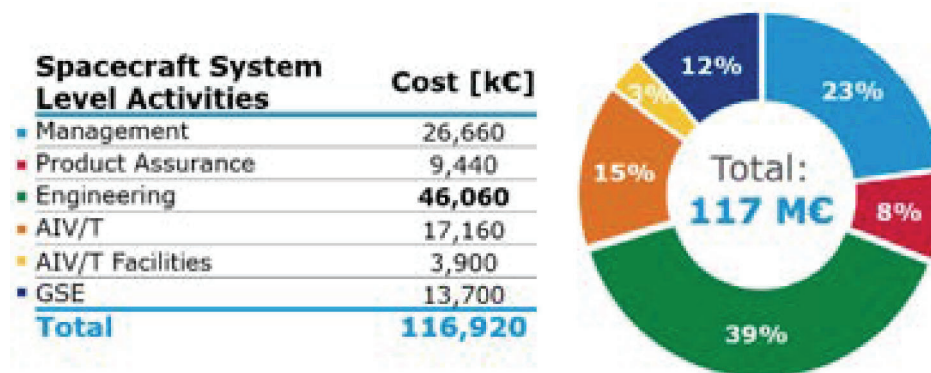


Fig. 44. Systems Level Activities Cost Breakdown

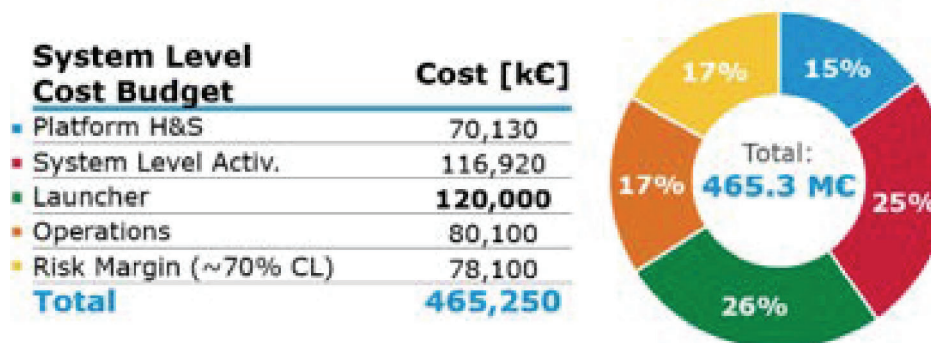


Fig. 45. Total Mission Budget Breakdown

Appendix L - Systems Engineering



Fig. 46. Mission Phases & System Modes

System/Domain Equipment	Number of Items	Unit Mass [kg]	Margin [%]	Mass [kg]
Spacecraft				2,149.66
Payload	1.00	1,000.00	0.00	1,000.00
Platform	1.00	1,149.66	0.00	1,149.66
AOGNC				52.74
COM				39.69
DH				6.04
PWR				178.79
PRO				328.99
STR				466.47
TC				76.94
Harness			5.00	113.14
Nominal Dry Mass at Launch				2,264.19
System Margin			20.00	229.93
Total Dry Mass at Launch				2,494.12
Pressurant mass		16.53	20.00	19.83
Propellant mass		3,680.22	2.00	3,753.83
Payload Launcher Adapter		85.00	-	85.00
Wet Mass				6,353

Fig. 47. Mass Budget

System/Domain/Equipment	LEOP	SAFE	COMMS_DTE	ECL	COMMS_PL	IDLE	MAN
Spacecraft	280.55	114.65	438.83	3,890.87	364.13	390.54	399.44
Payload	0.00	0.00	0.00	0.00	0.00	100.00	100.00
Platform	280.55	114.65	438.83	3,890.87	364.13	290.54	299.44
AOGNC	159.35	48.05	161.45	161.45	161.45	161.45	176.27
COM	0.00	0.00	137.28	52.08	52.08	4.08	4.08
DH	33.60	33.60	52.50	35.49	52.50	37.38	35.49
PWR	33.00	33.00	33.00	33.00	33.00	33.00	33.00
PRO	0.00	0.00	0.00	0.00	0.00	0.00	50.60
STR	54.60	0.00	54.60	0.00	65.10	54.63	0.00
TC	0.00	0.00	0.00	3,608.85	0.00	0.00	0.00
Totals w/o System Margin	280.55	114.65	438.83	3,890.87	364.13	390.54	399.44
System Margin	56.11	22.93	87.77	778.17	72.83	58.11	59.89
Totals with System Margin	336.66	137.58	526.60	4,669.04	436.96	448.65	459.33

Fig. 48. Power Budget

System/Domain	Number of Items	Unit Mass [kg]	Margin [%]	Iteration 0 - Top-down	Iteration 1 - 18.09	Iteration 2 - 19.09 AM	Iteration 3 - 19.09 PM
Spacecraft							
Payload							
INS	1.00	20.00	0.00	2,150.41	2,622.36	2,182.7	2,150.98
				1,000.00	1,000	1,000.00	1,000
Platform	1.00		0.00	1,000.00	1,000	1,000	1,000
AOGNC				68.83	38.44	49.44	52.74
COM				78.66	78.03	43.51	39.69
DH				29.50	14.04	6.04	6.04
MEC				39.33	0.00	0	0
PWR				216.32	304.47	180.91	178.79
PRO				127.82	288.45	328.99	328.99
STR				540.79	501.55	500.07	466.47
TC				49.16	37.38	73.74	76.94
				51.75	119.09	114.88	113.21
Harness		5.00					
Nominal Dry Mass at Launch				1,035.00	2,381.45	2,297.57	2,264.19
Platform System Margin		20.00		207.00	252.472	236.54	230.2
Total Dry Mass at Launch				1,242.00	2,633.92	2,534.11	2,494.39
Propellant mass				3,658.00	3,254.66	3,892.69	3,753.83
Pressurant mass					20.81	19.83	19.83
Wet Mass							
Launcher Adapter Ring				5,900.00	5,909.39	6,448.63	6,268.05
Launcher Mass				100.00	100	85	85
Launch Excess Margin				6,000.00	6,009.39	6,531.63	6,353.05
				13%	10%	5%	8%
A64 Performance				6,900.00	6,900	6,900	6,900

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Fig. 49. Spacecraft Evolution through Design Iterations

Appendix M - Misc. Figures

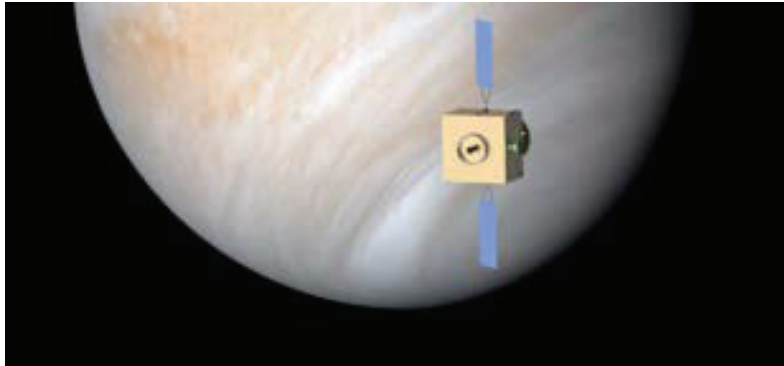


Fig. 50. Render #1 of the PRIMAVERA Spacecraft and the Atmospheric Probe arriving at Venus

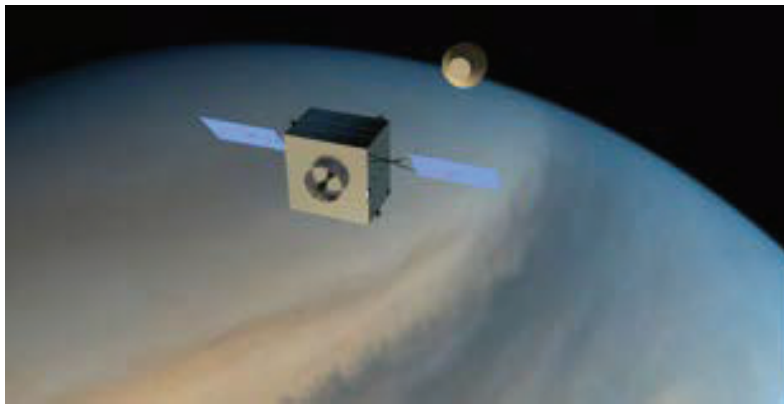


Fig. 51. Render #2 of the PRIMAVERA Spacecraft and the Atmospheric Probe following Deployment



Fig. 52. Render #3 of the PRIMAVERA Spacecraft and the Atmospheric Probe after Deployment

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